



AUTONOMOUS VEHICLES: LONGITUDINAL CONTROL USING FUZZY LOGIC

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**OTONOM ARAÇLAR: BULANIK MANTIK KULLANARAK BOYUNA
KONTROL**

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“Bu tezdeki tüm bilgilerin akademik kurallara ve etik ilkelere uygun olarak elde edildiğini ve sunulduğunu; ayrıca bu kuralların ve ilkelerin gerektirdiği şekilde, bu çalışmadan kaynaklanmayan bütün atıfları yaptığımı beyan ederim.”

Khaled HAMIDI

ÖZET

Lisans Tezi

OTONOM ARAÇLAR: BULANIK MANTIK KULLANARAK BOYUNA KONTROL

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Bu çalışma, otonom araçlarda boyuna kontrol sistemlerinin tasarımı ve optimizasyonunu incelemekte olup, bulanık mantık ve bulanık mantık denetleyicilerinin geleneksel PID denetleyicileri ile entegrasyonuna odaklanmaktadır. Araştırmanın amacı, dinamik ve karmaşık araç ortamlarını yönetebilecek ileri kontrol sistemleri geliştirerek otonom araçların güvenliğini, güvenilirliğini ve verimliliğini artırmaktır. Çalışmanın bir bölümü, bir prototip otonom araç modeli tasarlamayı içermektedir. Bulanık Mantık Denetleyici (BMD) ve hibrit BMD-PID kontrol sisteminin uygulanması ve test edilmesi, yanıt süresi ve kararlılıkta önemli iyileştirmeler göstererek, otonom araçlarda gerçek dünya uygulamaları için potansiyelini vurgulamaktadır.

Anahtar Sözcükler : Otonom Araçlar, Boyuna Kontrol, Bulanık Mantık, BMD, Kontrol Sistemleri, Prototip Tasarımı, Otonom Robot, Sistem Tanımlama, Diferansiyel Tahrikli Robot.

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ABSTRACT

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AUTONOMOUS VEHICLES: LONGITUDINAL CONTROL USING FUZZY LOGIC

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This study explores the design and optimization of longitudinal control systems in autonomous vehicles, focusing on the fuzzy logic and integration of fuzzy logic controllers with traditional PID controllers. The research aims to enhance the safety, reliability, and efficiency of AVs by developing advanced control systems capable of handling the dynamic and complex nature of vehicular environments. A part of the study involves designing a prototype model AV. The implementation and testing of the FLC and hybrid FLC-PID control system demonstrate significant improvements in response time and stability, highlighting its potential for real-world application in autonomous vehicles.

Key Word : Autonomous Vehicles, Longitudinal Control, Fuzzy Logic Controller, PID Controller, Control Systems, Prototype Design, Autonomous robot, System Identification, differential drive robot.

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TEŞEKKÜR

Bu zorluklar ve başarılarla dolu akademik yolculuğumun sonuna yaklaşırken, mezuniyet yolculuğumda bana destek olan ve yanımda duran herkese en içten teşekkürlerimi ve minnettarlığımı ifade etmek istiyorum. Öncelikle, her zaman ilk dayanağım olan sevgili aileme en derin teşekkürlerimi sunuyorum. Sürekli desteğiniz ve teşvikiniz, tüm engelleri ve zorlukları aşmamda itici güç oldu. Sizler her zaman ilham kaynağım ve gücüm oldunuz.

Değerli rehberlikleri ve kıymetli tavsiyeleri için danışman hocam Prof. Dr. Yusuf AYAN'e en içten teşekkürlerimi ve takdirlerimi sunmak istiyorum. Akademik desteği ve liderlik rolü, becerilerimi geliştirmemde ve bilgimi artırmamda büyük etki yarattı.

Değerli zamanlarını ve çabalarını çalışmamı değerlendirmek ve kabul etmek için harcayan kıymetli jüri üyelerine de teşekkür etmek istiyorum.

YTB ve Avrupa Birliği'nin bana verdiği bursların akademik ve araştırma hedeflerimi gerçekleştirmemde önemli rol oynadığını unutmam mümkün değil. Bana duyduğunuz güven ve cömert desteğiniz için teşekkür ederim.

Her zaman yanımda olan sevgili arkadaşım Dr. İbrahim SHAMTA'a ve bu yolculuk boyunca yanımda olan tüm arkadaşlarıma ve meslektaşlarıma teşekkür ederim. Desteğiniz ve teşvikiniz her zaman daha fazlasını başarmam için bir motivasyon kaynağı oldu. Birlikte geçirdiğimiz zamanlar ve değerli anılarımız her zaman kalbimde kalacak.

Bana ilim ve bilgi sunmak için ellerinden geleni yapan tüm değerli öğretmenlerime minnettarım. Özveriniz ve sadakatiniz, akademik ve mesleki gelişimimde derin bir etki yarattı. Hepinize teşekkür ederim.

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SİMGELER VE KISALTMALAR DİZİNİ

SİMGELER

e	: error (hata)
ce	: error derivative (hata türevi)
v	: voltage (gerilim)
A	: ampere (amper)
Θ	: angle (açı)
π	: ratio of circumference to diameter (çapın çevreye oranı)
t	: time (zaman)
f	: frequency (frekans)
R	: resistance (direnç)
I	: current (akım)
P	: power (güç)
E	: energy (enerji)
m	: mass (kütle)
d	: distance (mesafe)
v	: speed (hız)
T	: temperature (sıcaklık)
L	: inductance (endüktans)
C	: capacitance (kapasitans)

KISALTMALAR

- CAD : Computer-Aided Design (Bilgisayar Destekli Tasarım)
- DARPA: Defense Advanced Research Projects Agency (Savunma İleri Araştırma Projeleri Ajansı)
- DC : Direct Current (Doğru Akım)
- EMF : Electromagnetic Field (Elektromanyetik Alan)
- FLC : Fuzzy Logic Controller (Bulanık Mantık Kontrolörü)
- GPS : Global Positioning System (Küresel Konumlama Sistemi)
- LIDAR: Light Detection and Ranging (Işık Algılama ve Mesafe Ölçme)
- PI : Proportional-Integral (Oransal-Türevsel)
- PID : Proportional-Integral-Derivative (Oransal-Türevsel)
- PWM : Pulse Width Modulation (Darbe Genişlik Modülasyonu)
- RRT : Rapidly-exploring Random Tree (Hızlı Araştıran Rastgele Ağaç)
- RTOS : Real-Time Operating System (Gerçek Zamanlı İşletim Sistemi)
- SAE : Society of Automotive Engineers (Otomotiv Mühendisleri Derneği)
- SQP : Sequential Quadratic Programming (Sıralı Kare Programlama)

CHAPTER 1

INTRODUCTION AND OVERVIEW

1.1. Introduction

The advent of autonomous vehicles (AVs) marks a significant milestone in the evolution of transportation systems. These vehicles, capable of sensing their environment and navigating without human input, promise to enhance safety, efficiency, and convenience on roads worldwide. This research aims to explore the intricate aspects of AVs, with a particular focus on the design and implementation of a fuzzy logic controller for longitudinal control. The motivation for this study stems from the increasing demand for reliable and intelligent control systems in autonomous vehicles to ensure smooth and safe operation under varying conditions.

1.2. Background

The concept of autonomous vehicles has transitioned from science fiction to reality over the past few decades. Early developments in AV technology were characterized by experimental projects and theoretical research. Today, with advancements in sensor technology, artificial intelligence, and data processing, AVs are becoming more prevalent and sophisticated. Longitudinal control, which involves managing the vehicle's speed and maintaining safe distances from other vehicles, is a critical component of AV systems. Traditional control methods, such as PID controllers, have been widely used; however, they often face limitations in handling complex and nonlinear dynamic environments. This has led to the exploration of alternative approaches, including fuzzy logic controllers (FLCs), which offer greater flexibility and robustness in dealing with uncertainties.

1.3. Research Objectives

The primary objective of this research is to design and implement a fuzzy logic controller for the longitudinal control of an autonomous vehicle prototype. The specific objectives are as follows:

1. To review the current state of autonomous vehicle technology, focusing on the history, development, and classification of vehicle autonomy levels.
2. To analyze the key components and technologies involved in AVs, including sensory technologies, computational frameworks, and AI.
3. To design a prototype AV, incorporating a kinematic model and utilizing 3D printing for physical realization.
4. To develop a fuzzy logic controller, detailing the processes of fuzzification, rule base creation, inference, and defuzzification.
5. To integrate the FLC with a PID controller and evaluate the performance through modeling and simulation.
6. To implement the designed FLC in the AV prototype, validate its performance through experimental data, and compare it with traditional control methods.

1.4. Scope and Limitations

This research encompasses a comprehensive examination of autonomous vehicle technology and the development of control systems. The scope includes theoretical analysis, design, simulation, and practical implementation of control systems. However, the study is limited to longitudinal control and does not extensively cover lateral control or other aspects of AV operation, such as path planning and perception modules. Additionally, the prototype development is constrained by available resources and may not fully represent the capabilities of commercial AV systems.

1.5. Methodology

The research methodology involves a combination of literature review, theoretical modeling, simulation, and experimental validation. The following steps outline the approach:

1. Conduct an extensive literature review on autonomous vehicles, focusing on control systems and fuzzy logic applications.
2. Design a kinematic model for the AV prototype and utilize CAD software for 3D printing the physical model.
3. Develop the fuzzy logic controller, defining membership functions, rule bases, and inference mechanisms.
4. Integrate the FLC with traditional PID control systems and simulate the performance using software tools.
5. Implement the FLC in the AV prototype, collect sensor data, and analyze the control system's performance.
6. Validate the results through experimentation and compare the FLC's performance with that of PID and other controllers.

1.6. Organization of the Thesis

The thesis is organized into the following chapters:

- Chapter 2: Autonomous Vehicles - Provides an in-depth overview of AV technology, including definitions, history, development, and key components.
- Chapter 3: Design Prototype of an Autonomous Vehicle - Details the design and development of the AV prototype, including the kinematic model and 3D printing process.
- Chapter 4: Fuzzy Logic and Optimizing of Longitudinal Control - Explores the principles of fuzzy logic control and its application to longitudinal control in AVs.

- Chapter 5: Implementation of a Fuzzy Logic Controller for Longitudinal Control in an AV Prototype - Discusses the practical implementation, validation, and performance analysis of the FLC in the AV prototype.
- Chapter 6: Conclusion - Summarizes the findings, discusses the implications of the research, and suggests directions for future work.

This introductory chapter sets the stage for a detailed exploration of autonomous vehicle technology and the innovative control systems that drive their development. Through this research, it is anticipated that new insights and methodologies will be uncovered, contributing to the advancement of intelligent transportation systems.

CHAPTER 2

AUTONOMOUS VEHICLES

2.1. Introduction

The transportation sector is on the cusp of a paradigm shift driven by the emergence of Autonomous Vehicles (AVs). These self-driving vehicles hold the promise of fundamentally altering the way humans travel by removing them from the equation. This chapter embarks on a comprehensive exploration of AV technology, tracing its historical roots, analyzing its current state of development, and unveiling the potential it offers[1].

The concept of self-driving vehicles has captivated the imagination for decades, featuring prominently in science fiction and futuristic visions. Today, this vision is rapidly materializing due to advancements in artificial intelligence (AI), sensor technology, and computational power. The potential benefits of AVs are far-reaching, encompassing significant improvements in road safety, enhanced traffic flow, increased accessibility for individuals with mobility limitations, and a potential reduction in environmental impact[2].

However, widespread adoption of AVs faces significant challenges. Technical hurdles remain in areas of sensor perception, robust decision-making algorithms, and guaranteeing reliable operation in diverse and complex environments. Regulatory frameworks need to be established to address safety concerns, ethical considerations, and liability issues. Public perception and acceptance of this transformative shift in transportation necessitate careful consideration and education[2], [3].

This chapter serves as a springboard for a deep dive into AVs. It commences by establishing a clear definition of AVs and their varying levels of autonomy. A historical journey then unfolds, tracing the evolution of this technology from its conceptual origins to its current state of development. This historical perspective

provides crucial context for understanding the significant progress made and the challenges that lie ahead.

Following the historical context, a detailed industry overview is presented. This overview delves into the current landscape of AV development, identifying prominent companies, research institutions, and stakeholders at the forefront of this technological revolution. This analysis aids in understanding the diverse players driving innovation and shaping the future of AVs[3].

By establishing these introductory elements, a firm foundation is laid for further exploration. Subsequent sections of this chapter will dissect the technical underpinnings of AVs. The intricate interplay of sensor technologies like LiDAR, radar, and cameras, which allow AVs to perceive their surroundings, will be explored. An in-depth examination of the complex world of software and algorithms that process sensor data, enabling AVs to navigate, make decisions, and ultimately, operate autonomously, will then follow[4].

This exploration equips the reader with a thorough understanding of how AVs function, the technologies that power them, and the challenges that need to be overcome. With this comprehensive understanding, the chapter will then delve into the functional architecture of AVs, exploring the perception module, path planning strategies, and control systems that ensure safe and efficient operation[5].

Safety and security will be paramount throughout this exploration. The chapter will examine the rigorous safety protocols and measures being implemented to guarantee the reliable operation of AVs. Furthermore, it will analyze the inherent cybersecurity challenges in autonomous systems and the strategies devised to mitigate them[6].

In conclusion, this chapter provides a multifaceted exploration of AVs. By establishing a historical context, examining the current industry landscape, and diving deep into the technical aspects, a comprehensive understanding of this transformative technology is achieved. As the chapter navigates through the various components of AVs, the potential for a revolutionized transportation landscape becomes increasingly evident.

However, the challenges and opportunities that lie ahead are also brought to light, paving the way for a more informed and insightful discussion about the future of AVs.

2.2. Definition

Imagine a car that drives itself. This is the essence of AVs (AVs), also known as self-driving cars, driverless cars, self-piloting automobile or robo-cars. These vehicles represent a revolutionary leap in transportation technology, offering the ability to operate with varying degrees of human involvement.

The Society of Automotive Engineers (SAE) has established a six-level classification system to define these varying levels of autonomy, ranging from basic driver assistance features to full self-driving capability[7].

At its core, an AV is a complex system that relies on three key components:

- Sensing the World: AVs are equipped with a comprehensive sensor suite, like a digital nervous system. This suite typically includes LiDAR, radar, cameras, and ultrasonic sensors. These sensors work together to create a real-time picture of the environment, detecting other vehicles, pedestrians, traffic signals, and road infrastructure.
- Making Decisions: The raw data from the sensors is useless without powerful processing. Onboard computers, akin to a high-performance brain, analyze this sensory data in real-time. Sophisticated algorithms interpret the information, allowing the AV to understand its surroundings and make informed decisions[7].
- Taking Action: Once decisions are made, the AV translates them into actions through its control systems. These systems encompass steering, acceleration, and braking. By precisely controlling these functions, the AV navigates autonomously, adhering to traffic laws and ensuring passenger safety[7].

The potential benefits of AVs are vast. Imagine a world with:

- Safer Roads: Human error is a leading cause of accidents. AVs, by removing this factor, could significantly reduce traffic collisions and fatalities[7].
- Smoother Traffic Flow: Advanced communication capabilities in AVs could optimize traffic flow, resulting in reduced congestion and faster travel times[7].
- Inclusive Mobility: AVs could provide safe and reliable transportation options for individuals with mobility limitations or those in underserved areas.

- Cleaner Environment: Optimized traffic flow and the potential for electric AVs could contribute to a reduction in greenhouse gas emissions.

However, the road to widespread adoption of AVs is not without challenges. Technical hurdles like sensor refinement for diverse conditions and robust decision-making algorithms for complex situations need to be addressed. Clear regulations addressing safety, liability, and ethical considerations are crucial. Finally, gaining public trust in the safety and reliability of AVs is essential.

The journey towards self-driving vehicles is not new. The concept has captivated imaginations for decades, finding its place in science fiction and futuristic visions. To understand the current state of AV development and its future potential, let's delve into its history and development. This historical perspective will shed light on the significant progress made and the challenges that continue to be addressed.

2.3. History and Development

The concept of AVs has intrigued humans for centuries, demonstrating a longstanding fascination with mechanized transport. The earliest notions of automated mobility can be traced back to the Renaissance period, particularly with Leonardo da Vinci's design of a self-propelled cart in the late 15th century. This primitive vehicle, which was powered by coiled springs and capable of moving without human guidance, set the stage for the dream of autonomous machinery[3].

As the centuries progressed, this interest lay dormant until the advent of the automobile and subsequent technological advancements that could support automated driving. It was not until the 20th century that significant development began. In the 1920s and 1930s, experiments with radio-controlled cars showcased the potential for vehicles that could operate without direct human intervention[3].

The real momentum in AV development began in the 1980s with projects like the EUREKA Prometheus Project in Europe. This initiative focused on creating vehicles that could detect their lanes and navigate without human control using basic computer vision techniques. This project represented a crucial shift from theoretical research to practical application in AV technology[7].

The involvement of the U.S. Defense Advanced Research Projects Agency (DARPA) in the 1990s and 2000s further catalyzed advancements in this field. DARPA's Grand Challenges encouraged innovators to develop vehicles that could autonomously navigate both off-road and urban environments, spurring rapid developments in robotics, artificial intelligence, and sensor technology[7].

By the 2010s, major tech companies like Google, Tesla, and Uber had begun significant investments into AVs, testing them extensively on public roads and further refining the technologies necessary for safe and reliable automation. This period marked a surge in the development of sophisticated algorithms and improved sensor systems, propelling the industry toward the current state of near-functional AVs[1].

Today, the field of AVs is characterized by a blend of cutting-edge technology and ongoing challenges. The focus remains on enhancing sensor accuracy, improving decision-making algorithms for complex environments, and ensuring the safety and security of the systems. Despite these challenges, the evolution of AVs continues to progress, promising a future where transportation is not only automated but also safer, more efficient, and more accessible[1], [2].

2.4. Classification of Vehicle Autonomy Levels

The Society of Automotive Engineers (SAE) has established a widely recognized classification system that defines the different levels of vehicle autonomy [2]. This system ranges from Level 0, where a vehicle has no automation, to Level 5, where a vehicle can operate independently of a human driver under all conditions[5], [2], [8].

1. Level 0 (No Automation)

At this level, the vehicle has no automation features; all driving tasks are performed by the human driver. Example: Virtually all vehicles manufactured before the advent of electronic aids in driving fall into this category, such as the classic Ford Model T.

2. Level 1 (Driver Assistance)

This level allows the vehicle to control either steering or braking/accelerating but not both simultaneously. The human driver remains responsible for all other aspects of driving. Example: The Toyota Prius, introduced in 1997, was one of the first cars to include Adaptive Cruise Control, a key feature of Level 1 autonomy.

3. Level 2 (Partial Automation)

Vehicles at this level can simultaneously control both steering and braking/accelerating in specific circumstances. The human driver must remain engaged with the driving task and monitor the environment at all times. Example: The Mercedes-Benz S-Class (W220) introduced in 1998 featured Distronic Plus, a system that could control speed and the distance to the car in front, marking it as one of the first to offer such Level 2 capabilities.

4. Level 3 (Conditional Automation)

At Level 3, the vehicle can handle all aspects of the driving task in certain conditions, and the driver must be ready to intervene if necessary. Example: The Audi A8 (2017 model) was the first production car to feature Level 3 automation with its Traffic Jam Pilot system, allowing hands-free driving in certain highway traffic conditions up to 60 km/h.

5. Level 4 (High Automation)

This level signifies that the vehicle can perform all driving tasks independently in most conditions without any human intervention. Example: Google's Waymo project introduced its fully AVs in 2015, starting with modified Lexus SUVs and later developing their own prototype vehicles, making significant strides in Level 4 autonomy.

6. Level 5 (Full Automation)

At Level 5 autonomy, a vehicle is capable of performing all necessary driving operations in any conceivable driving scenario without the need for human intervention. This level represents the pinnacle of AV technology, where the car is fully independent and capable of navigating complex environments such as urban centers, highways, and rural roads under all weather conditions. As of now, no vehicles have

reached this stage of complete autonomy in a commercial capacity. However, numerous technology firms and automotive manufacturers are vigorously pursuing this goal. Leading companies in this field, such as Waymo and Cruise, are actively developing and refining the technologies required to achieve Level 5 autonomy. Their research focuses on enhancing machine learning algorithms, sensor technology, and data processing capabilities to ensure that vehicles can operate safely and efficiently without human oversight. This ongoing development underscores the significant potential and challenges of creating fully AVs that can adapt to the full range of real-world driving situations.

2.5. Driving Task

In AVs, the driving task involves complex operations categorized into three critical functions: perceiving the environment, planning navigation, and vehicle control. These functions utilize advanced sensor technologies and computational algorithms to ensure the AV operates safely and efficiently in diverse conditions[3].

2.5.1 Perceiving the Environment

This function is crucial as it involves the continuous acquisition and interpretation of external data through sensors. AVs are equipped with a comprehensive suite of sensors such as LiDAR, radar, cameras, and ultrasonic detectors. LiDAR sensors provide high-resolution, three-dimensional information about the vehicle's surroundings, essential for detecting obstacles and enabling precise navigation. Radar sensors complement this by measuring the velocity of objects, helping the vehicle assess its dynamic environment. Cameras add a layer of visual perception, critical for recognizing road signs, lane markings, and other navigation-related visuals. These sensors collectively create a detailed picture of the environment, which is vital for the next steps in the driving task[9], [10].

2.5.2. Planning Navigation:

Once the environment is perceived, AVs must plan their route from point A to point B. This involves complex decision-making algorithms that process the sensor data to

plot a course that avoids obstacles, adheres to traffic rules, and optimizes travel time. The planning algorithms evaluate multiple potential paths and make predictions about the movements of other road users to choose the safest and most efficient route. This function is continually updated in real-time as new data is received, allowing the AV to adapt to changing conditions and unexpected situations[11].

2.6. Sensory Technologies

AVs integrate a suite of advanced technologies that enable them to perceive, understand, and navigate their environments without human intervention. This section explores the critical technologies that form the backbone of AV functionality, categorizing them into sensory technologies, decision-making algorithms, and control systems.

The sensory capabilities of AVs are akin to human senses but are executed through technological means. These technologies include:

1. **LiDAR (Light Detection and Ranging)** is a foundational technology in AVs, offering high-resolution, three-dimensional insights into the vehicle's surroundings. This system works by emitting laser beams and timing how long they take to bounce back after hitting nearby objects, forming detailed point clouds that map the environment. This mapping is crucial for the precise detection of objects and obstacles, ensuring safe navigation. Although LiDAR has been traditionally costly and affected by environmental conditions like rain and fog, recent technological advances have significantly improved its accessibility and robustness for use in autonomous driving[10].
2. **Radar (Radio Detection and Ranging)**: Radar sensors are an essential component of the sensory suite in AVs, providing critical data on velocity and object detection that is indispensable for systems such as adaptive cruise control and collision avoidance. Radar operates by emitting radio waves and measuring the frequency shift of these waves when they bounce back from objects, a principle known as the Doppler effect. This allows radar to directly measure the speed of objects relative to the vehicle, offering robust functionality under various weather conditions, including fog, rain, or snow. Such capabilities enhance the vehicle's operational safety across diverse environments[9].
3. **Cameras**: Cameras are pivotal in the sensory array of AVs, providing vital visual information similar to human eyesight. Modern AVs employ multiple cameras to cover various angles, allowing them to effectively perceive road signs, detect lane markings, and monitor surrounding traffic.

4. Ultrasonic Sensors: These sensors use sound waves to detect the proximity of nearby objects, particularly useful for parking and low-speed maneuvers.
5. GPS and Inertial Measurement Units (IMUs): These technologies provide geospatial positioning and help with the vehicle's orientation, acceleration, and overall movement dynamics, crucial for route planning and tracking the vehicle's path[6].

2.7. Perception Module

The perception process in AVs involves several key elements, each critical to ensuring the effective and safe operation of the vehicle. These elements facilitate the vehicle's ability to understand and navigate its environment by processing and interpreting data from a variety of sensors. Here are the primary components of the perception process:

1. Road Surface Extraction: This involves identifying the boundaries and condition of the road. It is crucial for path planning and navigation, enabling the vehicle to determine where it is safe to drive.
2. On-road Object Detection: This includes the detection and classification of dynamic and static objects such as other vehicles, pedestrians, animals, barriers, and traffic infrastructure. This capability is essential for avoiding collisions and navigating complex traffic scenarios.
3. Object Tracking: Once objects are detected, tracking their movement over time helps predict their future positions, which is vital for dynamic path planning and collision avoidance.
4. Scene Segmentation: This process divides the visual input from cameras into distinct segments, such as road, sky, buildings, and vehicles. This helps in understanding the scene structure and in distinguishing between different types of objects and surfaces.
5. Signal and Sign Recognition: Recognizing traffic signals, signs, and lane markings is essential for compliance with traffic laws and for making appropriate driving decisions.
6. Anomaly Detection: Identifying unexpected features or conditions in the environment, such as road work, unexpected debris, or sudden changes in road conditions, which might not be typical or previously encountered.
7. Sensor Fusion: This crucial process combines data from various sensors like LIDAR, radar, cameras, and ultrasonic sensors to create a comprehensive and reliable perception of the environment. Sensor fusion enhances the accuracy and robustness of the perception system by mitigating the limitations of individual sensors.
8. Depth Perception: This involves calculating the distance to various objects and features in the environment, which is critical for maneuvering and path planning.

2.8. Object and Event Detection and Response (OEDR)

Object and Event Detection and Response (OEDR) is a pivotal system within AV technology, tasked with interpreting complex environmental data to ensure safe and efficient navigation. OEDR functions to accurately identify and respond to both static and dynamic objects and events in the vehicle's vicinity, such as other vehicles, pedestrians, and various roadway obstacles.

2.9. Path Planning

Path planning in AVs is a critical component that involves generating navigable paths from an initial point to a destination while avoiding obstacles, optimizing travel efficiency, and ensuring safety. This complex process replicates human decision-making in driving, integrating various technological advancements and algorithmic strategies to manage dynamic and potentially unpredictable environments. Below, the specialized and detailed aspects of path planning are explored.

2.10. Safety and Security

Ensuring safety and security in AVs is crucial due to the sophisticated technology and significant data they handle. This chapter explores the comprehensive safety protocols and cybersecurity measures essential for maintaining secure and reliable operations, backed by current research findings.

2.10.1. Safety Protocols and Measures to Ensure Secure Operations of Avs

Safety in AVs encompasses both physical safety measures and operational integrity protocols designed to handle various conditions and threats:

Functional Safety Standards: AVs adhere to international safety standards such as ISO 39003.

2.10.2. Cybersecurity Measures to Protect Against Digital Threats

The integration of advanced technologies in AVs makes them targets for cyber threats, necessitating robust cybersecurity measures:

Data Encryption: To safeguard the data integrity and privacy, strong encryption protocols are employed for data transmitted between the vehicle and external networks.

Intrusion Detection Systems (IDS): IDS are essential for real-time threat detection and management, helping to identify and mitigate potential attacks effectively.

Secure Software Lifecycle Management: This encompasses all stages of software development from design to deployment, ensuring that all software updates and patches address known vulnerabilities and enhance security measures.

Vehicle-to-Everything (V2X) Security: Security measures for V2X communications protect against unauthorized data access and manipulation, ensuring secure and reliable communication between vehicles and infrastructure.

Research highlights the importance of these cybersecurity measures in maintaining the safety and integrity of AV operations. For instance, the work by Kukkala, Thiruloga, and Pasricha (2022) emphasizes a roadmap for enhancing cybersecurity in AVs, addressing vulnerabilities through advanced artificial intelligence solutions. Additionally, Terlizzi, Mariani, and Glielmo (2021) discuss a Model Predictive Scheme to protect AVs from specific cyber-attacks like Replay Attacks and Denial of Service, which could compromise safety.

CHAPTER 3

DESIGN PROTOTYPE OF AN AUTONOMOUS VEHICLE

The primary purpose of the AV prototype is to facilitate a comprehensive study of autonomous navigation systems within a controlled and replicable environment. The prototype serves as a foundational platform to simulate and analyze various scenarios that AVs might encounter in real-world settings. This allows for iterative design improvements and the testing of hypotheses in a safe and cost-effective manner. By using a scaled-down model, researchers and developers can rapidly prototype different configurations of sensors and algorithms to evaluate their efficacy before implementing them in full-sized vehicles.

The applications of the AV prototype are diverse and extend across multiple sectors:

Educational Tool: The prototype can be utilized as a teaching aid in academic settings, particularly in engineering courses focused on robotics, artificial intelligence, and automotive technology. It provides students with hands-on experience in designing, programming, and troubleshooting autonomous systems.

Research and Development: In a research context, the prototype is instrumental in advancing the field of AVs. It enables researchers to test new algorithms for navigation, obstacle avoidance, and path planning. This is crucial for refining the technologies that will eventually be integrated into consumer and industrial AVs.

Industrial Testing: Companies involved in the development of autonomous driving technologies can use the prototype for proof-of-concept demonstrations and to validate the robustness of their designs in various operational environments. This helps in identifying potential issues and solutions early in the development process.

Simulation of Traffic Systems: The prototype can be employed to model and study traffic systems, providing insights into how AVs can be integrated into existing transportation networks. This can assist urban planners and policymakers in understanding the potential impacts of AVs on traffic flow, road safety, and urban infrastructure.

Custom Applications: Given its modular design, the prototype can be customized for specific applications such as delivery services, automated patrol vehicles, or environmental monitoring systems. Each application can be tailored to meet unique operational requirements, such as extended battery life for long-duration missions or specialized sensors for environmental data collection.

Overall, the prototype not only serves as a testbed for technological innovations but also as a versatile tool that can be adapted for various educational, research, and industrial applications. Its development represents a critical step towards understanding and solving the complex challenges associated with fully autonomous transportation.

3.1. Introduction

To design a model for an autonomous vehicle, it is essential to start by defining the purpose and required uses. The specific needs for the vehicle determine its design and requirements. A model has been presented that can be considered an autonomous vehicle, an autonomous robot, or a self-guided robot, which uses available and common market elements to create an autonomous vehicle.

The difference between a robot and AVs is that the term "vehicles" refers to robots that transport humans and can be more accurately described as self-driving cars. Since this does not apply to the proposed model, the term "autonomous robot" will be used.

Designing this robot offers a range of capabilities, providing a small-scale model for experimentation, which allows us to understand the mechanisms, technologies, and principles of AVs. Consequently, this model contributes to scientific knowledge through development and experimentation.

Developing and modeling this robot is not significantly different from larger AVs, and if the need arises to develop a vehicle, the same principles used in this model can be applied to that development.

Model capabilities:

- The model operates with both manual and automatic control systems, allowing control via a remote or computer.
- Energy studies ensure continuous operation for several hours without interruption.
- The model features the use of two types of batteries and the isolation of the computing unit from the motors, ensuring that voltage drops do not affect the computing unit, allowing the vehicle to operate smoothly and safely.
- The design is printable and can be assembled using M2, M3 screws.
- The design can be printed using a 3D printer with dimensions of 250x250, the most popular size for printers.
- With a camera, the robot can be seen and monitored while in motion.
- The robot includes all features of AVs, using navigation and global positioning systems for path planning with feedback from sensors, helping to avoid obstacles and move safely.

3.2. Kinematic Model

This study proposes the design of the vehicle such that each pair of motors operates on the same axis, making its movement similar to that of a tank. This design is known as a differential drive robot, and it resembles the movement of a wheeled robot with two controllable wheels. This type of design allows the vehicle to rotate around its axis and change direction with high precision, enhancing its ability to navigate complex environments and efficiently avoid obstacles.

The robot requires a linear velocity V and a heading angle θ . The robot's path can be accurately planned by controlling both the velocity and the orientation. The robot's motion is controlled by adjusting the velocities of the right (v_r) and left v_l wheels[16].

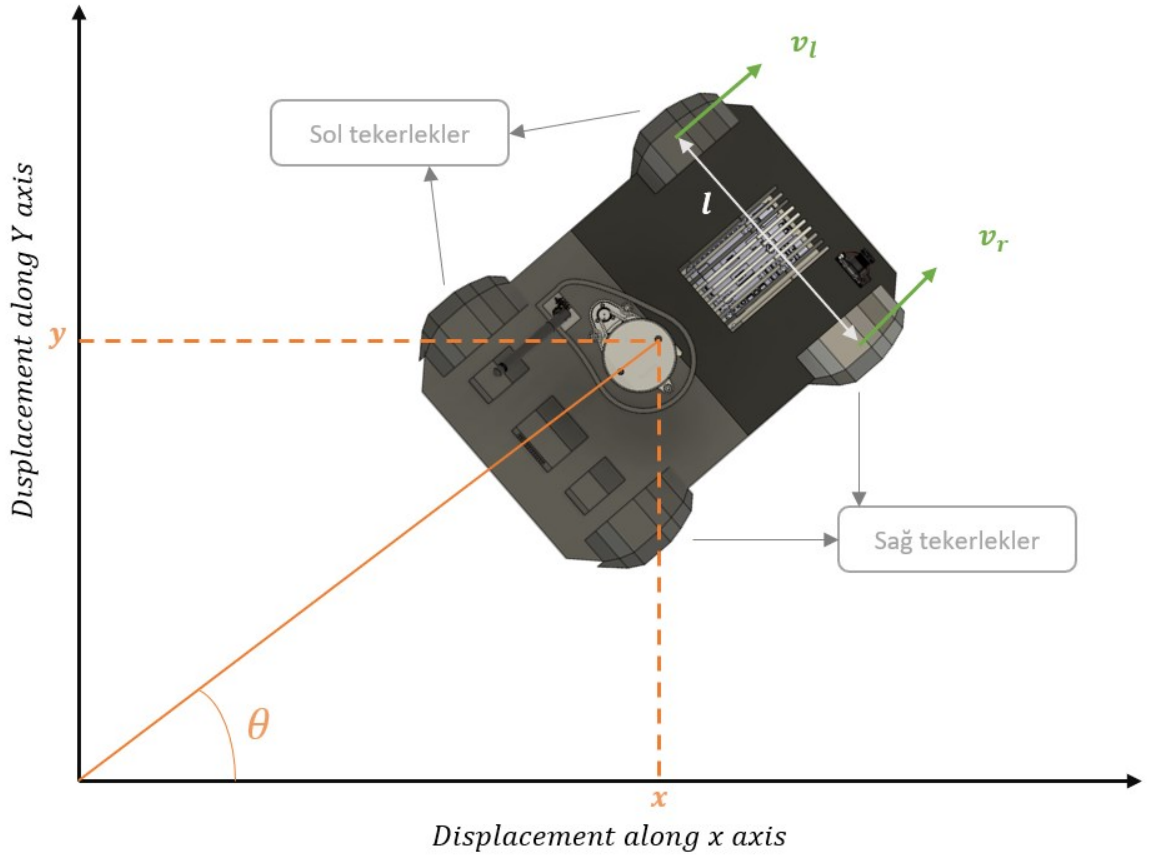


Figure 3.1 shows the differential drive robot Model

where V_r is the velocity of the right wheel,

- v_l is the velocity of left wheel,
- L is the distance between the center of the wheels,
- θ is the directional angle of the robot with respect to the reference coordinate system
- r is the radius of the wheel.
- The velocity of the robot is the average of the individual wheel velocities and is given by

$$v = \frac{v_r + v_l}{2} \quad (3.1)$$

And the Angular velocity of the robot is given by[17] :

$$\omega = \frac{v_r - v_l}{l} \quad (3.2)$$

The kinematic model of a differential drive robot describes how the robot's state variables—position and orientation—evolve based on its control inputs. The state of

the robot can be represented by its position (x, y) and its orientation (θ) . The control inputs are the linear velocity V and the angular velocity (ω) .

The kinematic equations can be expressed as[18]:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V \\ \omega \end{bmatrix} \quad (3.3)$$

Where:

- $(\dot{x})$ and $(\dot{y})$ are the rates of change of the robot's position in Cartesian coordinates.
- $(\dot{\theta})$ is the rate of change of the robot's orientation.
- (V) is the linear velocity of the robot.
- (ω) is the angular velocity of the robot.
- (θ) is the current orientation of the robot.

This model succinctly captures the relationship between the robot's motion in the plane and its control inputs. It is fundamental for understanding and implementing motion planning and control algorithms for differential drive robots. The model assumes that the robot moves without slipping, which induces a nonholonomic constraint that complicates the control problem but ensures realistic motion[16].

Substituting the linear velocity V from equation (3.1), in equations (3.2) and (3.3).

$$\dot{x} = \frac{V_r + V_l}{2} \cos \theta \quad (3.4)$$

$$\dot{y} = \frac{V_r + V_l}{2} \sin \theta \quad (3.5)$$

3.3. Model of a robot

(3.5)

In this project, the control of a differential drive robot's movement, specifically in terms of speed, braking, and acceleration, is achieved using suitable controllers that match the system's requirements. The modeling of the robot and the derivation of its transfer function are essential steps for creating these control systems, as the transfer function represents the vehicle's response to the desired speed[19].

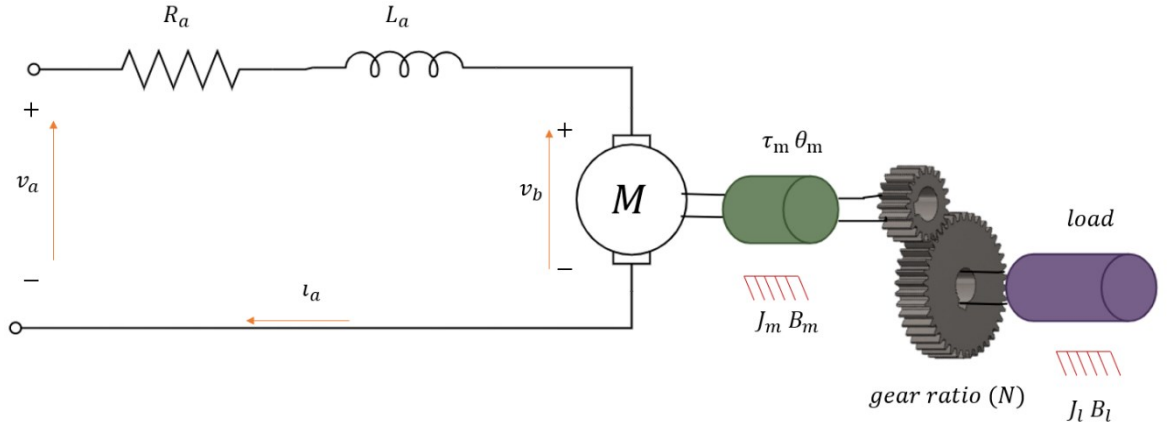


Figure 3.2 represents the equivalent model of a motor with the robot's weight coupled by gears.

The motor used in this study is a 6V DC motor with a 60:1 gear ratio, and its performance is influenced by the vehicle's weight. With four motors in the vehicle, the weight is distributed among them. The Kirchhoff's law of voltages for the rotor network, as described by equations (4) through (9), provides the basis for understanding the motor's electrical dynamics.

According to Kirchhoff's law, the voltage (v_a) across the motor is defined by[20]:

$$v_a = L_a \frac{di_a}{dt} + R_a i_a + v_b$$

The voltage (v_b) is proportional to the motor speed and is given by[16]:

$$v_b = k_b \omega_m$$

Applying the Laplace transform to these equations, assuming zero initial conditions, yields the armature current (I_a) as[17]:

$$I_a = \frac{V_a - V_b}{L_a s + R_a} = \frac{V_a - k_b s \theta_m}{L_a s + R_a}$$

The shaft torque (T_m), used for driving the load against the inertial and frictional torques, is defined by:

$$T_m = J_m \frac{d^2\theta_m}{dt^2} + B_m \frac{d\theta_m}{dt} + T_l = k_a i_a$$

Considering the gear ratio N , the equivalent inertia J_{eq} and damping B_{eq} are:

$$J_{eq} = J_m + \frac{J_L}{N^2}$$

$$B_{eq} = B_m + \frac{B_L}{N^2}$$

6. Substitute $I_a(s)$ into Torque Equation:

$$\left[k_a \left(\frac{V_a(s) - k_b s \theta_m(s)}{L_a s + R_a} \right) = J_{eq} s^2 \theta_m(s) + B_{eq} s \theta_m(s) \right]$$

Since $\Omega_m = s \theta_m(s)$:

$$k_a \left(\frac{V_a(s) - k_b s \theta_m(s)}{L_a s + R_a} \right) = J_{eq} s^2 \theta_m(s) + B_{eq} s \theta_m(s)$$

Simplify to find $\left(\frac{\theta_m(s)}{V_a(s)} \right)$:

$$\frac{k_a V_a(s)}{L_a s + R_a} - \frac{k_a k_b s \theta_m(s)}{L_a s + R_a} = J_{eq} s^2 \theta_m(s) + B_{eq} s \theta_m(s)$$

Combine terms involving $\theta_m(s)$:

$$\frac{k_a V_a(s)}{L_a s + R_a} = \theta_m(s) \left(J_{eq} s^2 + B_{eq} s + \frac{k_a k_b s}{L_a s + R_a} \right)$$

Final Transfer Function:

$$\frac{\theta_m(s)}{V_a(s)} = \frac{k_a}{(L_a s + R_a)(J_{eq} s^2 + B_{eq} s) + k_a k_b s}$$

The relationship between angular velocity and voltage, which will be used in our system, is as follows:

$$\begin{aligned}\frac{\dot{\theta}_m}{V_a} &= \frac{k_a}{(L_a + R_a/s)(J_{eq}s + B_{eq}) + k_a k_b} \\ &= \frac{k_a}{L_a J_{eq} s^2 + (L_a B_{eq} s + R_a J_{eq})s + (R_a B_{eq} + k_a k_b)}\end{aligned}$$

The relationship between motor angular velocity and gear box angular velocity is[21]:

$$\omega_m = \omega * N$$

So :

$$\frac{\omega}{V_a} = \frac{k_a/N}{L_a J_{eq} s^2 + (L_a B_{eq} s + R_a J_{eq})s + (R_a B_{eq} + k_a k_b)}$$

Where:

$$J_{eq} = J_m + \frac{J_L}{N^2}$$

$$B_{eq} = B_m + \frac{B_L}{N^2}$$

This transfer function describes the relationship between the motor's input voltage and its angular velocity. Considering the load effect coupled by gears to the motor spindle, the equivalent model (as shown in Figure 3.2) provides a comprehensive representation of the system dynamics. This detailed modeling and analysis are crucial for designing effective control systems that ensure precise and reliable operation of the differential drive robot.

When a DC motor lacks a datasheet, it becomes essential to identify its parameters through empirical means. Without the manufacturer's specifications, key parameters such as the motor's resistance, inductance, torque constant, back EMF constant, and mechanical characteristics must be experimentally determined. In Chapter 5, we will employ an experimental method to ascertain these parameters accurately. This

approach involves measuring the motor's response to various inputs and using these measurements to derive the transfer function. The transfer function derived from these experiments will reveal that the system has 0 zeros and 2 poles, which are crucial for understanding the motor's dynamics and for designing an appropriate control system.

zeros	poles
0	2

3.4. Design of the Robot

The design of the differential drive robot involves several crucial steps to ensure that the final product meets the desired specifications for autonomous movement and control. This section details the design process, including the use of design software, 3D printing, and considerations for component sizing and integration.

Component Sizing and Integration

An essential aspect of the design process is ensuring that all parts are appropriately sized and compatible with one another. This involves:

Motor Mounts: Designing motor mounts that securely hold the DC motors. The mounts must be robust enough to handle the torque generated by the motors while maintaining precise alignment.

- **Chassis:** Creating a chassis that houses all components, including the motors, battery, control electronics, and sensors. The chassis must be lightweight yet sturdy, providing a stable platform for the robot's operation (Figure 3.3).

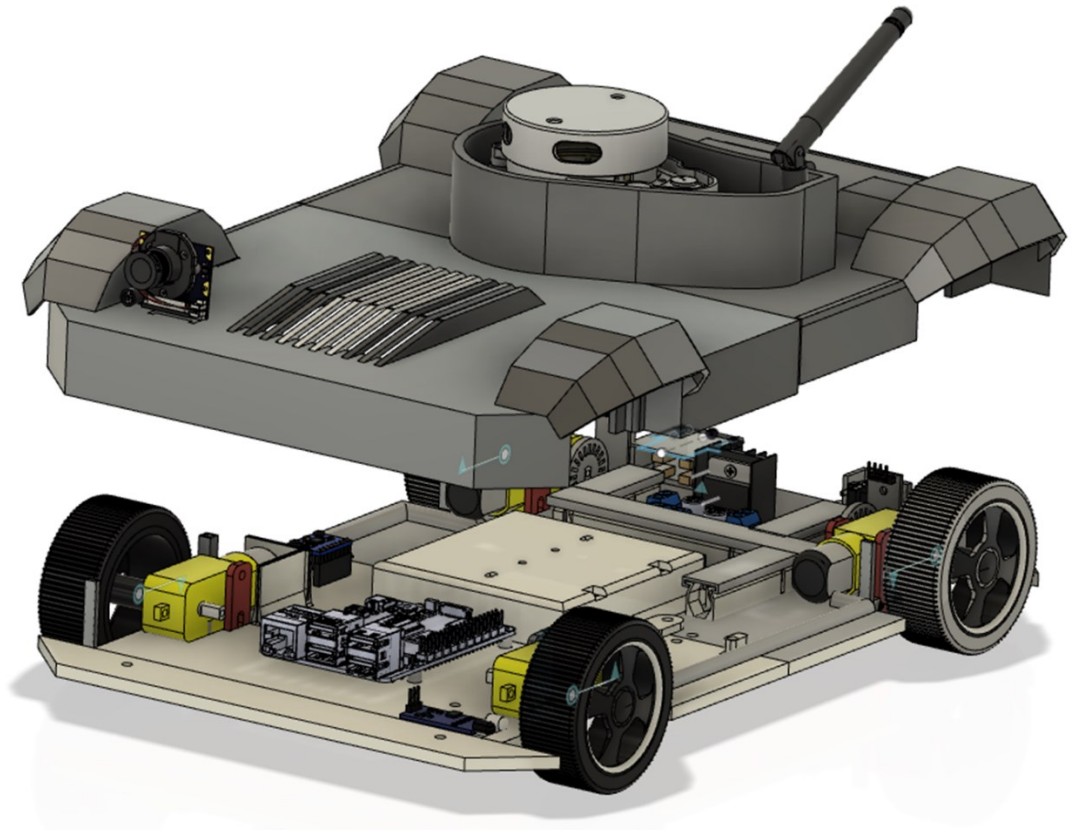


Figure 3.3. Overview of the robot chassis showing the arrangement of components.

- Battery Compartment: Including a compartment that safely secures the battery, ensuring easy access for replacement.
- Sensor Mounts: Adding mounts for sensors such as cameras, 3D lidar sensor, and other navigation aids. These mounts must position the sensors optimally to gather accurate data for the robot's control system.
- Cooling and Ventilation: Ensuring adequate cooling and ventilation for the Raspberry pi, preventing overheating and ensuring reliable operation.

Each part is designed with careful attention to size and fit, ensuring that all components work together harmoniously. The design is iteratively refined based on feedback from simulations and physical prototypes, leading to a well-integrated and functional robot. By using CAD software for design and a 3D printer for prototyping, the development process is streamlined, allowing for efficient creation and testing of the robot. This approach ensures that the final design is both functional and manufacturable, meeting the project's goals for a differential drive autonomous robot.

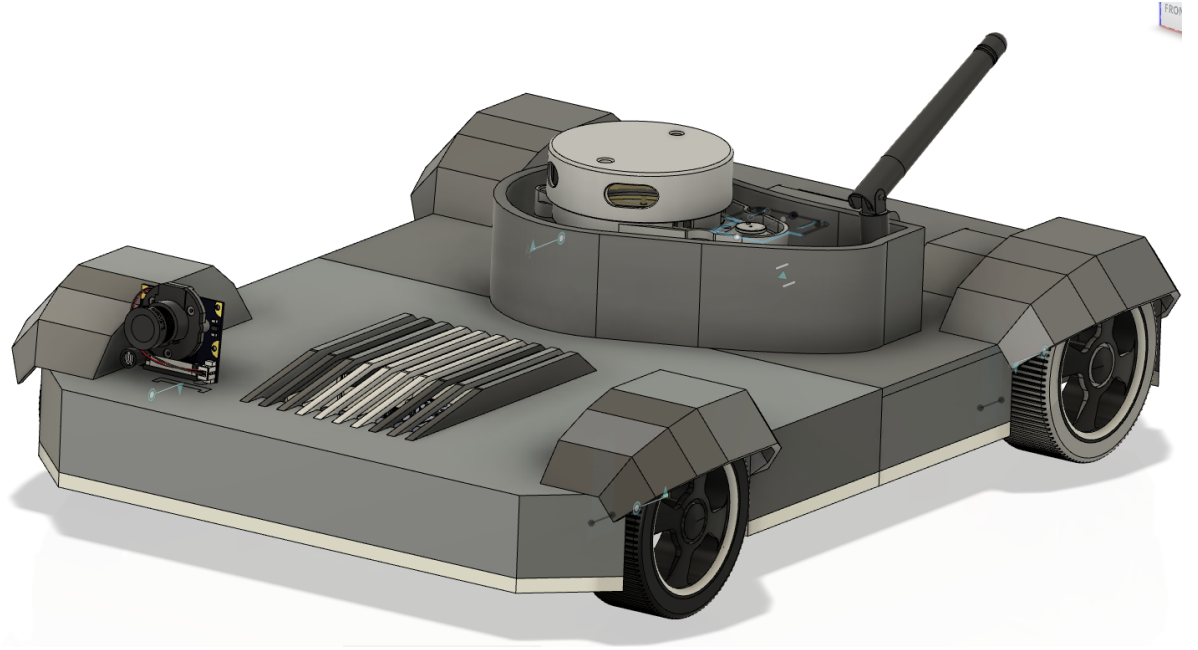


Figure 3.4. the proposed design is shown.

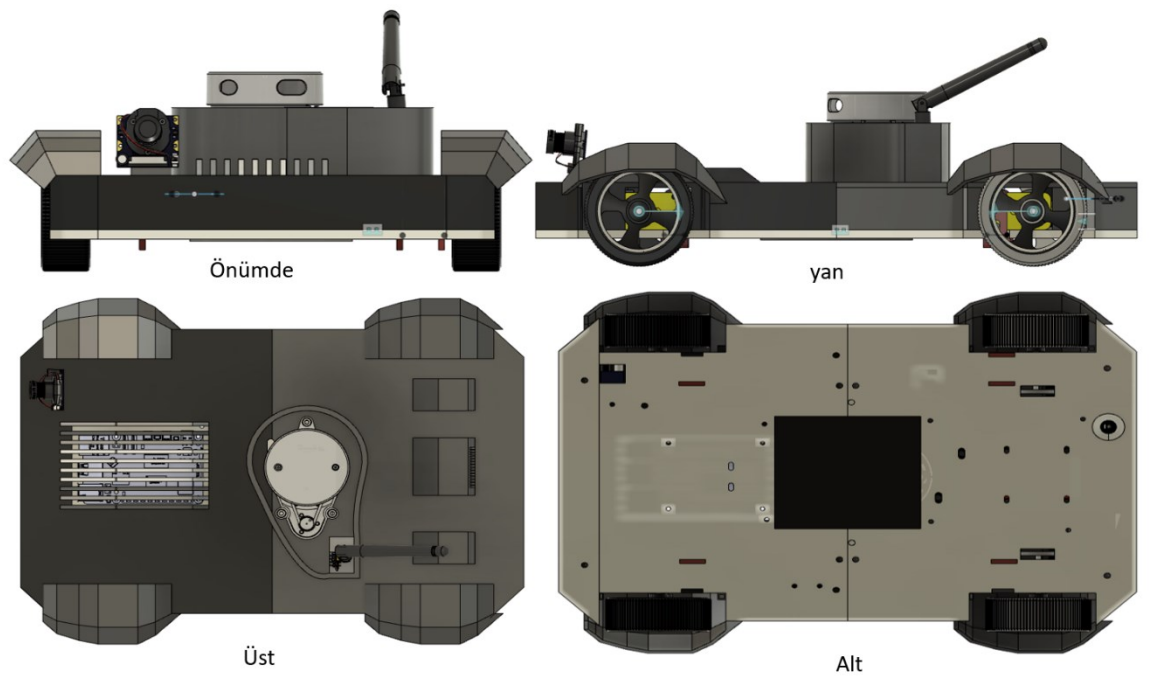


Figure 3.5. Multi-View Diagram of the Robot Design

CHAPTER 4

FUZZY LOGIC AND OPTIMIZING OF LONGITUDINAL CONTROL

4.1. Introduction

Longitudinal control is a fundamental aspect of vehicle dynamics, focusing on the acceleration and deceleration processes that govern speed management and safe following distances[19]. In the context of AVs, optimizing longitudinal control is critical for ensuring safety, efficiency, and passenger comfort. Traditional control systems, particularly Proportional-Integral-Derivative (PID) controllers, have long been the backbone of such automated control processes due to their simplicity and effectiveness in various conditions. However, as AVs operate in increasingly complex environments, the limitations of PID controllers, such as handling nonlinear systems and reacting to dynamic changes in the driving environment, become apparent[14].

Fuzzy logic, an approach that allows for more adaptive decision-making through handling imprecise information, presents a promising alternative. By mimicking human reasoning and accommodating the uncertainties inherent in real-world driving, fuzzy logic can potentially offer more nuanced and flexible control strategies[22], [23], [24]. This chapter explores and compares these two methodologies—PID and fuzzy logic—in optimizing longitudinal control systems in AVs[14], [25]. It assesses their efficacy, application, and potential integration to highlight how they can enhance the autonomous driving experience, ultimately contributing to the development of safer and more reliable vehicles. Through this analysis, the chapter provides insights into the evolution of control systems in automotive technology, emphasizing innovations that could shape the future of autonomous driving.

4.2. Overview of Longitudinal Control Systems

Longitudinal control systems are crucial components in the automation of vehicle dynamics, focusing primarily on managing the forward and reverse movements of vehicles. This type of control is vital for maintaining and adjusting the speed of a vehicle as well as ensuring appropriate spacing between vehicles in traffic. In AVs, effective longitudinal control is essential for safety, efficiency, and comfort.

The basic premise of longitudinal control is to regulate the acceleration and braking forces in response to various driving scenarios. This is typically achieved through two main strategies: cruise control and adaptive cruise control. Traditional cruise control maintains a vehicle's speed at a driver-set level until deactivated or overridden by the driver pressing the brake. In contrast, adaptive cruise control (ACC) adjusts the vehicle's speed automatically to maintain a safe distance from vehicles ahead, using radar or lidar sensors to detect the distance and relative speed of other vehicles on the road[25].

In AVs, longitudinal control systems are more complex and are integrated with a multitude of sensors and data sources to make real-time decisions. These systems utilize a variety of inputs, including vehicle speed, throttle position, brake pressure, and environmental data from onboard cameras and sensors. The primary objective is not only to maintain a steady speed or safe following distance but also to optimize travel time, reduce fuel consumption, and enhance passenger comfort[14].

With the introduction of more sophisticated technologies, control strategies have evolved from rule-based algorithms to more advanced approaches such as fuzzy logic, neural networks, and deep learning. These methods offer significant improvements over traditional control systems, especially in terms of handling the uncertainties and dynamic environments associated with driving scenarios. FLCs, for instance, excel in managing imprecise inputs and nonlinear systems, making them particularly suitable for the complex decision-making processes in AV control[14].

4.3. PID Controllers

Proportional-Integral-Derivative (PID) controllers are one of the most prevalent and effective types of feedback controllers used across various industries, including automotive, aerospace, manufacturing, and electronics[26]. Their popularity stems from their simplicity and robust performance in a wide range of operating conditions. PID controllers are particularly crucial in systems where continuous control is needed to regulate key variables such as speed, temperature, pressure, or other measurable factors[13].

4.3.1. Basic Principles of PID Control

A PID controller calculates an "error" value as the difference between a desired setpoint and a measured process variable. The controller attempts to minimize this error by adjusting the process control inputs. The control effort from a PID controller is a sum of three terms: proportional (P), integral (I), and derivative (D)[13], [26].

1. Proportional: The proportional term produces an output that is proportional to the current error value. The proportional response can be adjusted by multiplying the error by a proportional gain, K_p . A higher value will increase the system's response speed but can also lead to an overshoot where the process variable exceeds the setpoint.

$$y(t) = K_p e(t) = K_p (u(t) - y(t - 1))$$

2. Integral: The integral term is concerned with the accumulation of past errors. If the applied force is not enough to bring the error to zero, this term integrates the error over time, thereby increasing the controller action until the error is nullified. The integral action is scaled by the integral gain, K_i , and helps eliminate the steady-state error in the system[26].

$$y(t) = K_i \int_0^t e(t) dt$$

3. Derivative: The derivative term is a prediction of future error, based on its rate of change. It provides a damping force that reduces the magnitude and speed of the overshoot caused by the proportional and integral components. The derivative action is scaled by the derivative gain, K_d , and helps improve the stability and speed of the system's response[27].

$$y(t) = K_d \frac{de(t)}{dt}$$

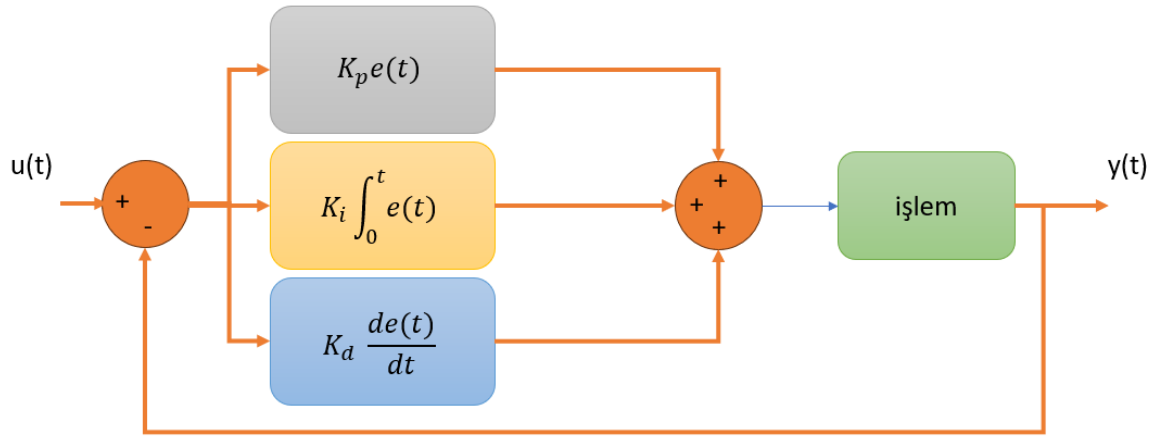


Figure 4.1. PID Kontrolör Şeması

4.3.2. Advantages and Limitations

Advantages:

- Simplicity: PID controllers are straightforward to implement and tune.
- Efficiency: They are computationally light, making them suitable for real-time applications.
- Adaptability: They can be fine-tuned with adjustable parameters to meet various operating conditions.

Limitations:

- Performance: PID controllers may not perform well with non-linear systems unless carefully tuned.
- Sensitivity to Noise: The derivative term can amplify noise in the measured process variable, potentially leading to erratic control signals.
- Tuning Complexity: Proper tuning of PID parameters K_p , K_i and K_d are crucial and can be challenging, especially in systems with interacting dynamics[26].

4.4. Fuzzy Control System

Fuzzy logic control (FLC) represents a paradigm shift in control engineering. Traditionally, control systems were based on mathematical models that required precise inputs and used well-defined linear equations for processing. FLC, however, introduces an approach based on "degrees of truth" rather than the usual "true or false" (binary) reasoning associated with classical logic[22], [24]. This technique simulates human decision-making processes and deals effectively with uncertain or imprecise information[28].

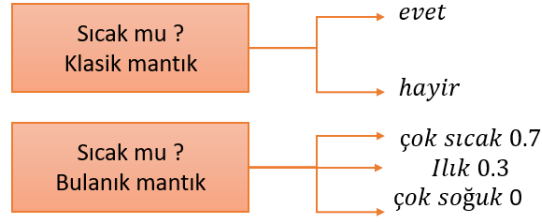


Figure 4.2. Comparison of Classical Logic and Fuzzy Logic for Temperature Evaluation

The core of FLC lies in its use of fuzzy sets, developed by Lotfi Zadeh in the mid-1960s, to handle the concept of partial truth. FLC systems convert input data into logical variables that can take on continuous values between 0 and 1, rather than discrete, binary choices. This allows for a more nuanced approach to control that can mimic the vagaries of human logic and reasoning[29].

Applications of FLC are vast and varied, including everything from consumer products like cameras and washing machines to complex industrial systems and AV navigation. The adaptability and simplicity of FLCs make them particularly useful in systems where a mathematical model of the control process is difficult to obtain or the system behavior is highly uncertain[22], [24].



Şekil 4.3. Doğruluk ihtiyacı ile önem arasındaki farkı gösterir

The development and implementation of fuzzy logic control have thus made significant contributions to both theoretical and practical aspects of control engineering, offering a flexible and robust alternative to traditional control methods.

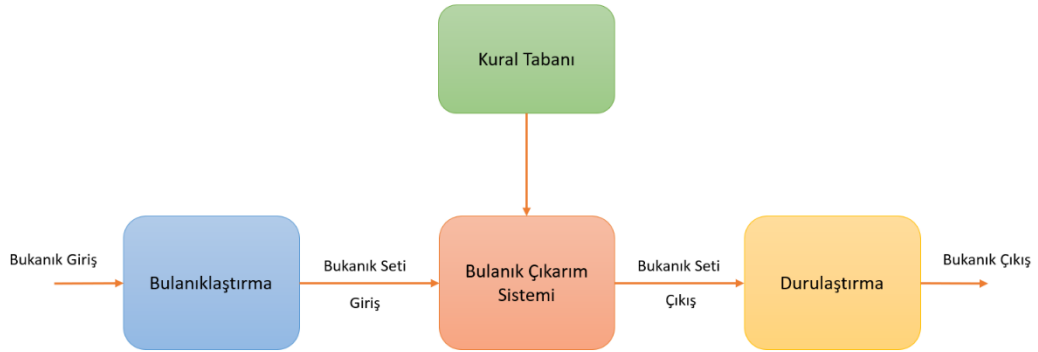


Figure 4.4. Fuzzy Logic Architecture

4.4.1. Fuzzification

Fuzzification step helps to convert inputs. It allows to convert, crisp numbers into fuzzy sets. Crisp inputs measured by sensors and passed into the control system for further processing. Like Room temperature, pressure, etc[22].

4.4.2. Rule Base

It contains all the rules and the if-then conditions offered by the experts to control the decision-making system. The recent update in fuzzy theory provides various methods for the design and tuning of fuzzy controllers. This updates significantly reduce the number of the fuzzy set of rules[22].

4.4.3. Inference Engine

It helps you to determines the degree of match between fuzzy input and the rules. Based on the % match, it determines which rules need implment according to the given input field. After this, the applied rules are combined to develop the control actions[30].

4.4.4. Defuzzification

At last, the Defuzzification process is performed to convert the fuzzy sets into a crisp value. There are many types of techniques available, so you need to select it which is best suited when it is used with an expert system[22].

4.5. Fuzzification Module

The first step in a fuzzy inference system where crisp numerical inputs (real-world numerical values) are converted into fuzzy values. This step is essential for the system to handle imprecision and ambiguity in decision-making using fuzzy logic[22].

In fuzzification, each input variable is assessed against one or more membership functions to determine the degree to which these inputs belong to each of the appropriate fuzzy sets. Membership functions can take various forms, such as triangular, trapezoidal, Gaussian, etc., and they assign a membership value between 0 and 1 to each input[24].

figure 4.5. provides a depiction of two mutually exclusive membership functions labeled as "Soğuk" and "Sıcak". Each function is depicted as a distinct block that spans a defined range on a temperature scale, demonstrating a partitioning where "Soğuk" encompasses the interval from 0° to 20° Celsius, while "Sıcak" covers the 20° to 40° Celsius range. This configuration exemplifies a crisp partitioning method where each temperature value is strictly classified as either "Cold" or "Hot," with no overlap between the two sets[24].

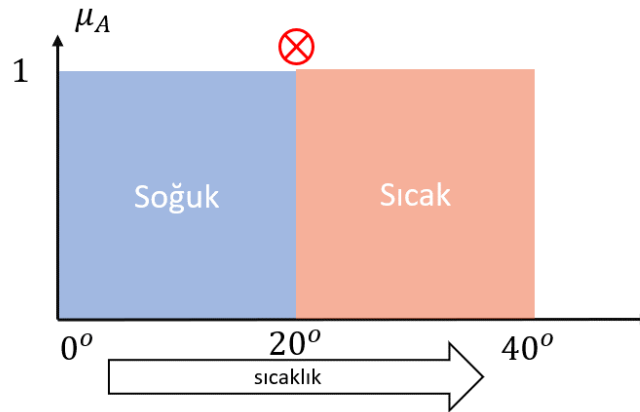


Figure 4.5. Classical logic

In contrast, figure 4.6. illustrates a more sophisticated scenario with overlapping membership functions for "Soğuk", "Ilık", and "Sıcak". This arrangement allows for a single temperature value to possess partial membership in multiple sets concurrently. For example, a temperature around 20° Celsius could be considered both "Cold" and "Warm" to varying degrees. This overlap enables a nuanced representation that more accurately mirrors real-world perceptions and conditions[24].

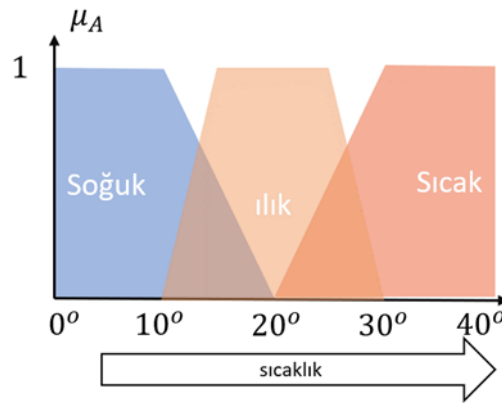


Figure 4.6. Fuzzy logic

The choice of fuzzification strategy—non-overlapping versus overlapping sets—significantly influences the decision-making dynamics within a fuzzy logic control system:

This strategy offers a more flexible control mechanism, where inputs can simultaneously activate multiple rules to varying extents. This capability is particularly beneficial in complex systems where nuanced and adaptive responses are crucial[22], [24].

4.6. Fuzzy Logic Membership Functions

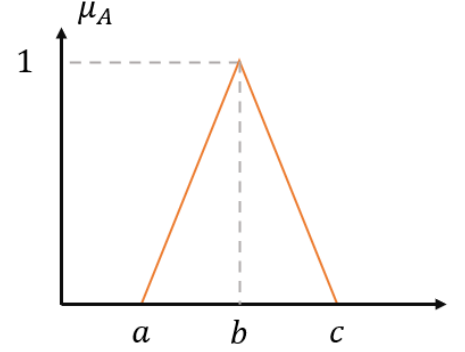
Fuzzy logic uses different types of membership functions to define how each point in the input space is mapped to a membership value between 0 and 1. These functions are crucial for determining how "true" an input is in relation to a fuzzy set. Here's a detailed look at the common types of membership functions used in fuzzy logic:

4.6.1. Triangular Membership Function

The triangular membership function is shaped like a triangle, defined by three points a , b , and c . The function's value is zero at a and c , and one at b [23].

Equation:

$$\mu_A(x) = \text{triangle}(x, a, b, c) = \begin{cases} 0 & x < a \\ \frac{(x-a)}{(b-a)} & a \leq x \leq b \\ \frac{(c-x)}{(c-b)} & b \leq x \leq c \\ 0 & c \leq x \end{cases}$$



$$\mu_A(x) = \max\left(0, \min\left(\frac{(x-a)}{(b-a)}, \frac{(c-x)}{(c-b)}\right)\right)$$

Example: If the temperature is 18°C, and the fuzzy sets for temperature are defined as "Cold" (0-15°C), "Moderate" (10-20°C), and "Warm" (15-25°C), the fuzzification process might determine that 18°C has a membership value of 0.6 in "Moderate" and 0.4 in "Warm", and 0 in "Cold".

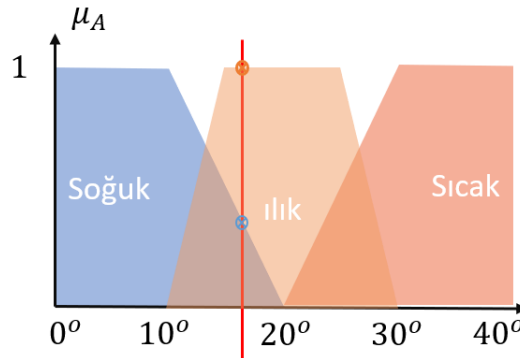


Figure 4.7. Fuzzy Logic Membership Functions for 17°C

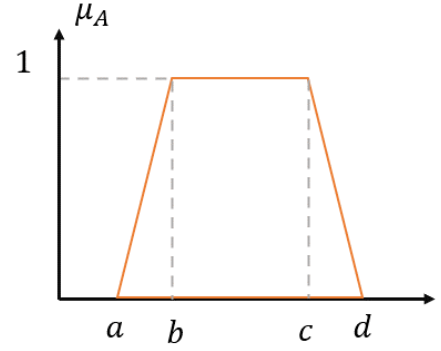
4.6.2. Trapezoidal Membership Function

The trapezoidal membership function is defined by four points: a, b, c, and d, resembling a trapezoid. The function's value is flat and equal to 1 between b and c[22].

Equation:

$$\mu_A(x) = \text{trapezoid}(x; a, b, c, d) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & b \leq x \leq c \\ \frac{d-x}{d-c} & c \leq x \leq d \\ 0 & d \leq x \end{cases}$$

$$\mu_A(x) = \max\left(0, \min\left(\frac{(x-a)}{(b-a)}, 1, \frac{(d-x)}{(d-c)}\right)\right)$$

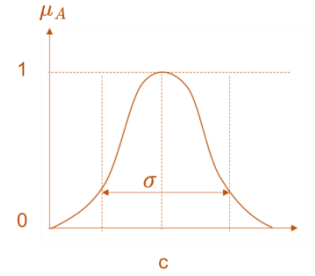


4.6.3. Gaussian Membership Function

The Gaussian membership function is shaped like a bell curve, defined by a center c and a width parameter σ [29], [30].

Equation:

$$\mu_A(x) = e^{-\frac{(x-c)^2}{\sigma^2}}$$



In fuzzy logic systems, especially those involved in control systems or decision-making processes, integrating multiple membership functions becomes essential for capturing the nuances of real-world scenarios. For instance, in the example shows in Figure 4.7; consider a temperature control system where the temperature input must be categorized into several linguistic variables such as "Cold", "Warm," and "Hot." Each of these categories can be represented by different types of membership functions.

The integration of these various membership functions within a single system allows for a more detailed and accurate modeling of input variables. For example, in an environmental control system, different sensors might provide inputs that need to be fuzzified using different MFs based on their sensitivity and the nature of data they collect. By employing multiple MFs, the system can accurately interpret complex and often overlapping sensory data into actionable fuzzy sets that drive the control rules.

The choice of membership functions and their integration must be tailored to the specific requirements of the application, balancing precision, computational efficiency, and the ability to handle real-world ambiguity. The use of multiple membership functions helps to enhance the robustness and flexibility of fuzzy logic systems, making them more adept at handling diverse and variable input scenarios. the

use of symmetric membership functions is essential for modeling and analyzing variables that exhibit both positive and negative values. These functions are typically named to reflect the degree and direction of the variable they represent. Labels such as NB (Negative Big), NS (Negative Small), ZE (Zero), PS (Positive Small), and PB (Positive Big) are commonly used to categorize inputs according to their linguistic significance. This method allows for a nuanced interpretation of data, making it possible to handle complex scenarios where the input may vary significantly across a spectrum.

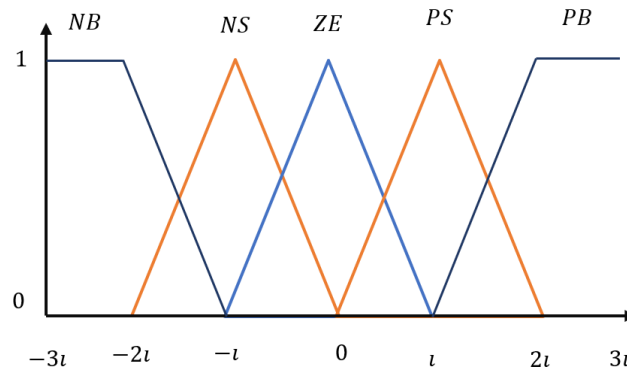


Figure 4.8. Symmetric Membership Functions for Fuzzy Logic Analysis.

As shown in Figure 4.8, the membership functions are plotted to represent different levels of an input variable. NB and PB extend flatly, indicating strong membership in the respective categories over a broader range, while NS, ZE, and PS peak sharply at specific points, facilitating a clear differentiation between smaller variations around the center. This layout is crucial for systems requiring precise control and gradation, such as temperature regulation or speed control in engineering applications.

4.7. Fuzzy Logic Logical Operations

Fuzzy Logic Logical Operations involve manipulating fuzzy sets using operations such as AND, OR, and NOT, which accommodate varying degrees of membership[29].

4.7.1. Fuzzy AND (Intersection)

The AND operation in fuzzy logic is implemented using the main function, which takes the minimum of the membership values of two fuzzy sets[22].

Equation:

$$\mu_{A \cap B} = \mu_A(x) \text{ and } \mu_B(x) = \min(\mu_A(x), \mu_B(x))$$

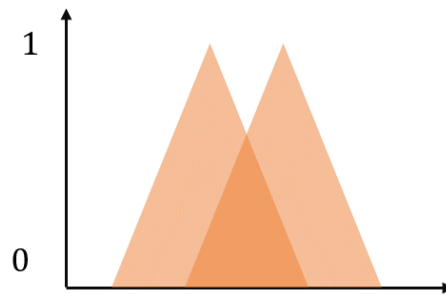


Figure 4.9. AND operation

4.7.2. Fuzzy OR (Union)

The OR operation in fuzzy logic uses the max function, returning the maximum of the membership values of two fuzzy sets[24].

Equation:

$$\mu_{A \cup B} = \mu_A(x) \text{ or } \mu_B(x) = \max(\mu_A(x), \mu_B(x))$$

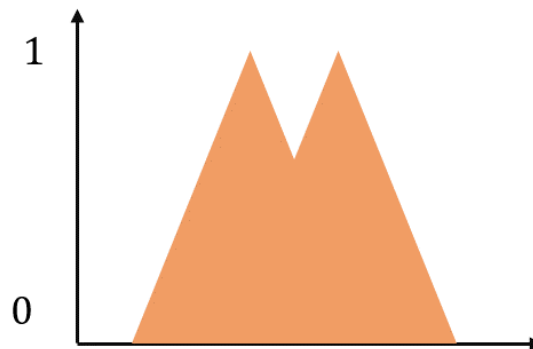


Figure 4.10. OR operation

4.7.3. Fuzzy NOT (Complement)

The NOT operation computes the complement of a fuzzy set by subtracting the membership value from 1.

Equation:

$$\mu_{\bar{A}} = \text{not } \mu_A(x) = 1 - \mu_A(x)$$

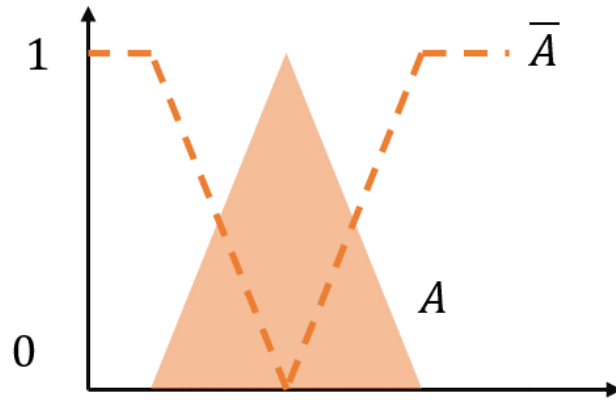


Figure 4.11. Complementation process.

4.8. Rule Base in FLCs

At the core of any FLC lies its rule base, a structured set of fuzzy IF-THEN rules. These rules are pivotal in translating linguistic inputs into actionable outputs, encompassing the expert knowledge or empirical data about the system's behavior. Each rule links conditions, defined by fuzzy sets and their associated membership functions, with a consequent action or state[30].

The logical operators within these rules determine how inputs are combined and interpreted to produce decisions. Two primary operators are used:

T-norms (AND operations): These operators are used to intersect fuzzy sets, thereby integrating multiple condition statements. A common t-norm is the Minimum operator, which for two conditions A and B with membership values a and b respectively, is defined as[30]:

$$T(a, b) = \min(a, b)$$

T-conorms (OR operations): These operators handle the union of fuzzy conditions, useful when any of several conditions could trigger the same output. The Maximum operator is a typical t-conorm, defined for conditions A and B as[24]:

$$S(a, b) = \max(a, b)$$

In FLCs, the interaction and aggregation of rules are essential for handling complex scenarios where multiple conditions may be partly true simultaneously. The inference mechanism processes each rule based on the current inputs and uses t-norms to evaluate the degree to which each rule applies. The results from individual rules are then aggregated using a t-conorm, which synthesizes the outputs into a cohesive action.

For instance, consider a heating system with the following rules:

1. IF Temperature is Very Low THEN Heating is High
2. IF Temperature is Low THEN Heating is Medium

If the temperature is between 'Very Low' and 'Low', both rules may apply to some extent. The output for the heating level would be aggregated from the outputs of these two rules to decide the final command for the heater.

When there is more than one input, rules are used to compare each two values, and the rules are given in the form of a table. The number of rules is calculated by[22]:

$$(5 \text{ kural})^{(2 \text{ giris})} = 25 \text{ kural}$$

Table 4.1. shows the rules of the system according to the input values.

	NB	NS	ZE	PS	PB
NB	NB	NB	NB	NS	ZE
NS	NB	NS	NS	ZE	PS
ZE	NB	NS	ZE	PS	PB
PS	NS	ZE	PS	PS	PB
PB	ZE	PS	PB	PB	PB

Each cell in the table corresponds to a specific fuzzy logic rule, which determines the output based on combinations of input linguistic variables. These variables typically represent states or measurements that affect the control decision, such as distance to an obstacle, relative velocity, or directional alignment.

In this particular table, the columns and rows are labeled with linguistic terms which could, for instance, denote error and change.

These terms could represent, for example, deviation from a lane center (error) and rate of change of this deviation (derivative of error) in an AV navigation scenario.

An example:

IF the Error in Lateral Position is NS (small negative deviation to the left) AND the Rate of Change of Lateral Position is PS (moving towards the center but slowly), THEN the Steering Adjustment is ZE (no adjustment needed).

4.9. Defuzzification

Defuzzification is the process of converting a fuzzy output set into a single crisp output, which is essential for the fuzzy logic system to interact with real-world applications. After the fuzzy inference engine has applied all the fuzzy rules to the input data and aggregated the outputs, defuzzification provides a definitive action or value based on these fuzzy conclusions. And the purpose of defuzzification is to allow fuzzy systems to make practical decisions in environments that require precise, scalar outputs. It ensures that the inherently vague and overlapping conclusions of a fuzzy logic system can be translated into specific actions or measurable quantities[22], [24].

Common Defuzzification Methods:

There are several methods used to defuzzify the output of a fuzzy logic system, each with its specific approach and application suitability:

4.9.1. Centroid Method (Center of Area or Gravity)

- The most widely used method due to its intuitive appeal and fairness[23].
- It computes the center of gravity of the area under the curve of the aggregated fuzzy set, providing a balance point that represents the output.
- Formula: $COA = \frac{\sum_{i=1}^n \mu_i x_i}{\sum_{i=1}^n \mu_i}$
- Where μ_i is the membership value at a discrete point x_i .

4.9.2. Mean of Maximum (MOM)

- Identifies the average of the maxima of the aggregated fuzzy set[30].
- This method is simpler than the centroid but can be less sensitive to changes in the shape of the fuzzy set[30].

4.9.3. Smallest of Maximum and Largest of Maximum

- Chooses the smallest or largest value among the maximum membership values[23].
- Useful in systems where extreme values are more significant[23].

4.10. Fuzzy Inference

Fuzzy inference (FI) is the process by which a fuzzy logic system deduces a response from a set of inputs using fuzzy logic. This core mechanism of a FLCler employs a

series of logical steps to simulate human decision-making based on vague or imprecise information[23].

Fuzzy inference combines fuzzy logic principles with an established rule base to produce an output that reflects the nuances of real-world situations. This process allows for approximate reasoning by evaluating the degree to which input variables satisfy the conditions of each rule. The FI process typically involves the Rule Evaluation

1. fuzzification of the input variables
2. rule evaluation
3. aggregation of the rule outputs
4. defuzzification.

Once the firing strengths for all rules are determined, they are combined to form a single fuzzy set for each output variable. This aggregation step typically uses the max operator to ensure that the strongest rule (highest firing strength) dictates the output response[23].

Implication Method:

A method that modifies the output membership function based on the firing strength of the rule. Common methods include clipping the output membership function at the height of the rule's firing strength or scaling the output membership function by the firing strength.

There are primarily two types of fuzzy inference systems, each with its unique approach and applications:

4.10.1. Mamdani-Type Fuzzy Inference

The output of each rule is a fuzzy set. It is more intuitive and widely used, especially in consumer products and simple control systems[24].

The final output is obtained by defuzzifying the aggregated fuzzy output set[24].

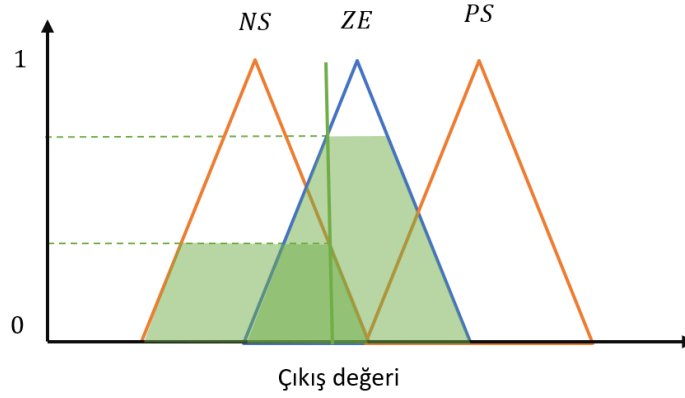


Figure 4.12. Defuzzification Process Using the Centroid Method.

4.10.2. Sugeno-Type Fuzzy Inference

Introduced in 1985 [16], the Sugeno method shares many similarities with the Mamdani method. The initial steps of the fuzzy inference process—fuzzifying the inputs and applying the fuzzy operator—are identical in both methods. The key difference lies in the Sugeno method's use of linear or constant output membership functions. This approach results in a final output that is a weighted average, eliminating the need for defuzzification and thereby enhancing computational efficiency[24].

A typical rule in a Sugeno fuzzy model has the form:

If Input 1 = x **and** Input 2 = y , **then** Output $z = ax + by + c$

The final output of the system is the weighted average of all rule outputs, computed as[24]:

$$\text{Output} = \frac{\sum_i w_i z_i}{\sum_i w_i}$$

where w_i are the weights determined by the membership functions, and z_i are the rule outputs.

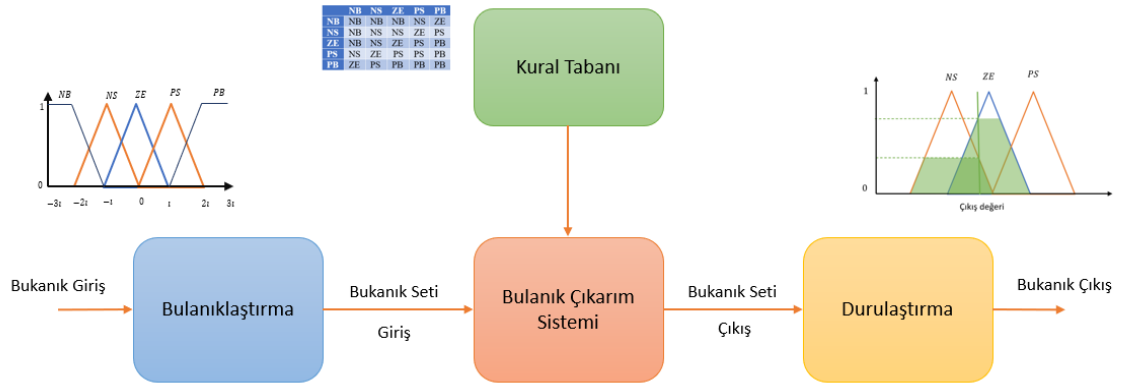


Figure 4.13. Overview of FLC System Flowchart.

4.11. Integration with PID Controllers

Integrating FLCs with PID controllers offers a robust solution to managing the complexities of longitudinal control:

- **Enhanced Adaptability:** The FLC component provides high-level decision-making capability that adapts to the qualitative aspects of the driving environment, which are difficult to quantify precisely[24], [31].
- **Precision Control:** The PID controller can take the output of the FLC and fine-tune it, providing precise control actions necessary for smooth and safe vehicle operations. This integration is particularly beneficial in scenarios where both rapid response and fine adjustments are critical.
- **Hybrid Control Architecture:** In a typical setup, the fuzzy controller might determine the necessary acceleration or braking force level, and the PID controller could then adjust the actual pedal or brake application to achieve these levels as smoothly as possible.

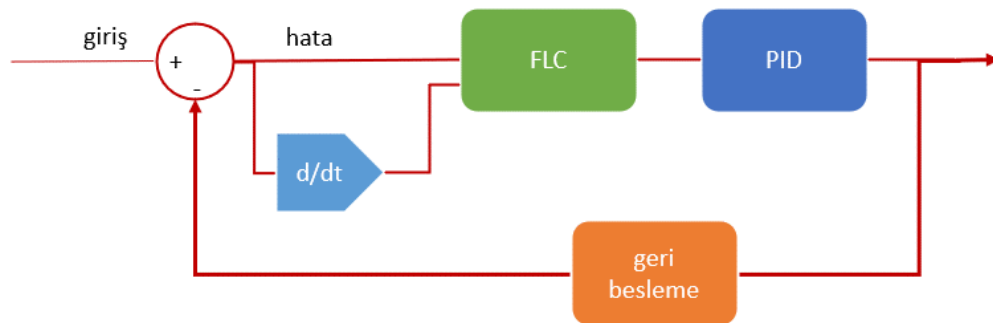


Figure 4.14. FPID kontrolörü.

CHAPTER 5

IMPLEMENTATION OF A FUZZY LOGIC CONTROLLER FOR LONGITUDINAL CONTROL IN AN AV PROTOTYPE

5.1. Introduction

In this chapter, the primary objectives are to develop a precise mathematical model and to design effective controllers for the autonomous vehicle system. Building upon the foundational work presented in previous chapters, this chapter delves into the empirical extraction of the vehicle's motor transfer functions, the processing and analysis of sensor data, and the formulation of sophisticated control strategies.

In Section 5.2, the focus is placed on the empirical methods used to derive the transfer functions for each motor. These functions are crucial for understanding the vehicle's dynamics, especially when manufacturer-specified characteristics are unavailable. MATLAB's System Identification Toolbox is utilized to generate approximate transfer functions based on empirical data. The accuracy of these models is ensured through comprehensive data collection and meticulous experimental setups. Initially, the vehicle is operated, and its response to varying input speeds is recorded, capturing acceleration and reaction data.

The methodology for capturing and processing sensor data is detailed in subsequent sections. A speed measurement system, utilizing a rotary encoder attached to one of the vehicle's wheels, is implemented. This encoder generates pulses as the wheel rotates, each pulse corresponding to a fraction of a wheel revolution. The angular speed is calculated by measuring the time between consecutive pulses, which is then converted to linear speed. The collected data is systematically recorded in a text file, which includes the target speed and the actual speed of the vehicle at fixed intervals.

Section 5.3 introduces the design of various controllers, including a Fuzzy Logic Controller (FLC), a traditional PID controller, and a hybrid Fuzzy-PID (FPID)

controller. Each controller's design process is meticulously detailed. The FLC is developed using a Mamdani inference system with two inputs and one output, governed by 25 rules that determine the control actions. The PID controller is designed to serve as a baseline for performance comparison. The FPID controller integrates fuzzy logic principles with the traditional PID control strategy, aiming to combine the strengths of both approaches.

The chapter culminates in a comprehensive discussion of the results, where the performance of the different controllers is rigorously compared. Their responses to varying inputs are evaluated to identify the strengths and limitations of each control strategy. The performance metrics, including rise time, overshoot, settling time, and absolute error, are analyzed to provide valuable insights into the practical applications of each controller.

Through this chapter, a clear understanding of the system's dynamics is established, and control strategies that enhance the vehicle's stability, responsiveness, and overall performance are developed.

5.2. Extract the transport function

In the third Chapter, the transfer function for each motor of the vehicle was deduced, encompassing a set of motor characteristics typically specified in the datasheet provided by the manufacturer. However, commonly used motors in the market often lack these specified characteristics, necessitating the use of empirical methods to extract them.

MATLAB's System Identification Toolbox offers tools that utilize empirical data to derive an approximate transfer function of the motor. The accuracy of the derived function improves with the richness of the data and the precision of the experimental setup. Initially, it is essential to operate the vehicle and record its response to varying input speeds, capturing acceleration and reaction data[32].

This is accomplished through a specifically designed program that controls the vehicle operation based on user inputs, with the vehicle's responses logged in a .txt format log

file. This file includes two columns: the first column records the user-inputted target speed, and the second column records the actual speed of the vehicle. Sampling is set at a fixed interval (sample time), valued at 0.05 seconds.

The operation of the speed sensor is based on the principles of motion and light detection. As the motor spins, a recurring light signal is emitted from a light source mounted on the axle. This signal is captured by a sensor positioned near the motor. The ring mounted on the axle features multiple slots. When the light encounters a slot, part of it reflects back to the sensor, creating a light pulse. The speed of the motor's rotation is determined by counting the number of pulses received by the sensor within a specific time period.

The ring used in this model contains 20 slots, meaning the sensor detects 20 pulses per full revolution. The speed calculation involves measuring the time elapsed between each pair of pulses using the formula:

$$T = \frac{1}{20\Delta t} \text{ rev/sec}$$

To convert this to angular velocity:

$$\text{angular velocity} : \omega = \frac{2\pi}{T} \text{ rad/s}$$

The input signal (desired speed) includes step changes, ramps up, and ramps down.

Figure 5.1. illustrates the applied signal:

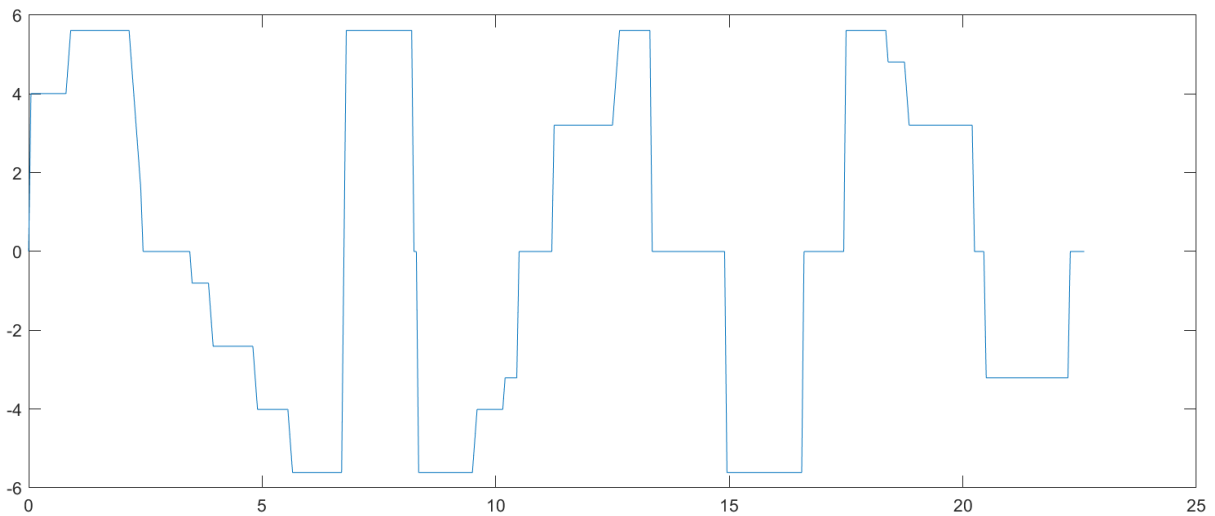


Figure 5.1. Illustrates the input signal (angular speed) over time with a sampling period of 0.05 seconds.

To accurately monitor and control the robot's motion, we utilize a sophisticated Python-based schema for data acquisition.

5.2.1. Reading Sensor Data

The robot uses a rotary encoder attached to one of its wheels to measure how fast it is turning. This encoder generates pulses as the wheel rotates. Each pulse corresponds to a small fraction of a wheel revolution. The number of pulses per full revolution is defined by `PULSES_PER_REVOLUTION`.

The `on_inturrupt()` function is triggered each time the encoder detects a pulse (using the `Button` class from `gpiozero`, where the encoder is set up on pin 24). This function calculates the time between consecutive pulses (`last_pulse_duration`) and uses this to compute the wheel's angular speed (`last_angular_speed`). The angular speed is then smoothed using a simple filter that blends the current and previous speed readings to avoid abrupt changes.

```
52 def on_inturrupt():
53     global last_pulse_duration, pulse_start_time, last_angular_speed, lspeed
54     if pulse_start_time is not None:
55         pulse_end_time = time()
56         last_pulse_duration = pulse_end_time - pulse_start_time
57         lspeed = last_angular_speed
58         last_angular_speed = 1.0/(last_pulse_duration*20.0) # rev/s
59         last_angular_speed = 0.95*last_angular_speed + 0.5*lspeed
60         if(last_angular_speed) > 5:
61             last_angular_speed = lspeed
62         get_current_speed()
63     pulse_start_time = pulse_end_time
64
65 # Attach event handlers to the encoder
66 encoder.when_pressed = on_inturrupt
--
```

Figure 5.2. Reading Sensor Data python code.

5.2.2. Converting Sensor Data to Speed

Once the angular speed is determined, the `get_current_speed()` function converts this angular speed to linear speed in millimeters per second, based on the wheel's circumference (`wheel_circumference`). This linear speed is then capped at `max_speed` to prevent unrealistic speed values.

```
def get_current_speed():
    global last_angular_speed, wheel_circumference, last_speed
    if last_angular_speed is None:
        return 0

    speed = (last_angular_speed * wheel_circumference)
    if speed < 0 or speed > max_speed:
        last_speed = -1
    last_speed = speed
```

Figure 5.3. Code of Converting Sensor Data to Speed.

5.2.3. Recording Data in a Text File

The robot's speed and other telemetry data are continuously recorded in a log file. This is achieved by a loop inside a try block that listens for keyboard commands (using the keyboard library) to control the robot's movements. These movements are then correlated with the target speed (target_speed) and the last computed angular speed (last_angular_speed).

```
try:
    with open(f"log_{datetime.now().strftime('%H_%M_%S')}.txt", "a") as file:
        while True:
            file.write(f"{target_speed}\t{last_angular_speed}\n")
            file.flush() # Ensure the data is written to the file
            sleep(0.05) # Reduce sensitivity and CPU usage
except KeyboardInterrupt:
    motor_left.stop()
    motor_right.stop()
    print("Program terminated")
```

Figure 5.4. code to save data as a text file.

Whenever the robot's speed or direction is adjusted using inputs, these changes along with the current angular speed are logged in a tab-separated format in a text file named with the current time to ensure that each session has a unique log file. The file.write() statement records these values, and file.flush() ensures the data is immediately written to the file, preventing data loss in the event of a program interruption.

5.2.4. transport function extraction

20	0	0
21	0	0
22	0	0
23	0	0
24	0	0
25	0	0
26	2.4	0
27	2.8	0
28	3,0	0
29	3.2	0.04060048868265584
30	3.2	1.1412440011413216
31	3.2	1.812267120059336
32	3.2	2.2787651380751326
33	3.2	2.5728643218714646
34	3.2	2.7569627372700163
35	3.2	2.8556204777510326

Figure 5.5: the saved data

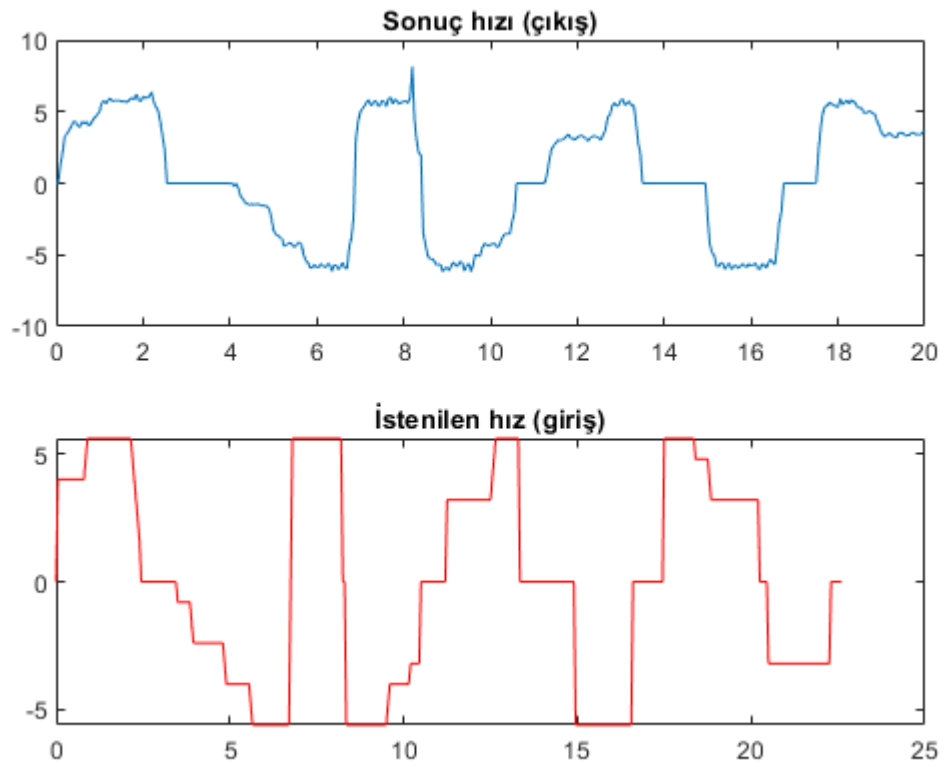


Figure 5.6. Comparison of Desired Input Speed and Actual Output Speed Over Time.

In MATLAB, the file was opened using the `readtable` function, which conveniently converts the tabulated data into a MATLAB table format, facilitating easy

manipulation and analysis. The data was then visualized using MATLAB's plotting functions to create a comprehensive graph that illustrated the relationship between the input commands and the vehicle's actual performance over time.

To develop a mathematical model of our robotic system, we employed system identification techniques, leveraging the data collected from controlled experiments as depicted in Figure 7. The primary objective was to accurately derive the transfer function that models the relationship between the input and output speeds of the system.

1. Data Preparation

The experimental data, comprising desired input speeds and corresponding actual output speeds, were systematically logged and formatted into a dataset suitable for system identification analysis. This dataset captured the system's dynamic response over time to varying input commands, providing a rich basis for model estimation.

2. Model Selection and Estimation

Given the nature of our system, a transfer function model was deemed appropriate for capturing the dynamics. Using MATLAB's System Identification Toolbox, we initially experimented with various model orders to ascertain the best fit. The model estimation process was facilitated by the `tfest` function, which provided a flexible framework for refining our model structure.

We explored several optimization algorithms during the transfer function estimation:

1. Least Squares Method (`lsqnonlin`):

Adjusted for function tolerance to ensure precision.

```
Options = tfestOptions;  
Options.SearchMethod = 'lsqnonlin';  
Options.SearchOptions.FunctionTolerance = 1e-05;  
tf1 = tfest(data, 2, 0, Options);  
  
compare(data, sys);
```

2. Adaptive Gauss-Newton line search (Wills-Ninness):

Provided robust estimation with a fit of 88.71% to the experimental data.

3. Interior-Point and Sequential Quadratic Programming (SQP):

Both methods confirmed consistent fit results of 88.71%, demonstrating the reliability of our model under various optimization conditions.

```
>> sys

sys =
  From input "u1" to output "y1":
      514.1
  -----
  s^2 + 49.38 s + 508.3

Name: sys
Continuous-time identified transfer function.

Parameterization:
  Number of poles: 2   Number of zeros: 0
  Number of free coefficients: 3
  Use "tfdata", "getpvec", "getcov" for parameters and their uncertainties.

Status:
  Estimated using TFEST on time domain data "data".
  Fit to estimation data: 88.5% (stability enforced)
  FPE: 0.1577, MSE: 0.1552
```

Figure 5.7. : Shows the extracted transfer function and accurac

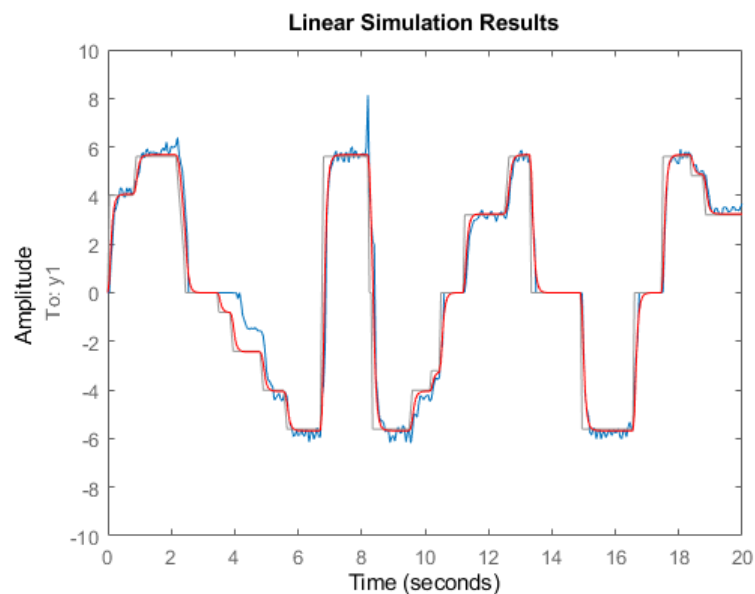


Figure 5.8. shows the output response of the produced TF and the measured angular velocity

5.2.5. Validation and Results

The estimated model was rigorously validated against the experimental data to verify its accuracy. ensuring that the model not only fit the historical data but also reliably predicted future system behavior. The compare function in MATLAB provided a fit percentage of 51.09%, indicating moderate accuracy.

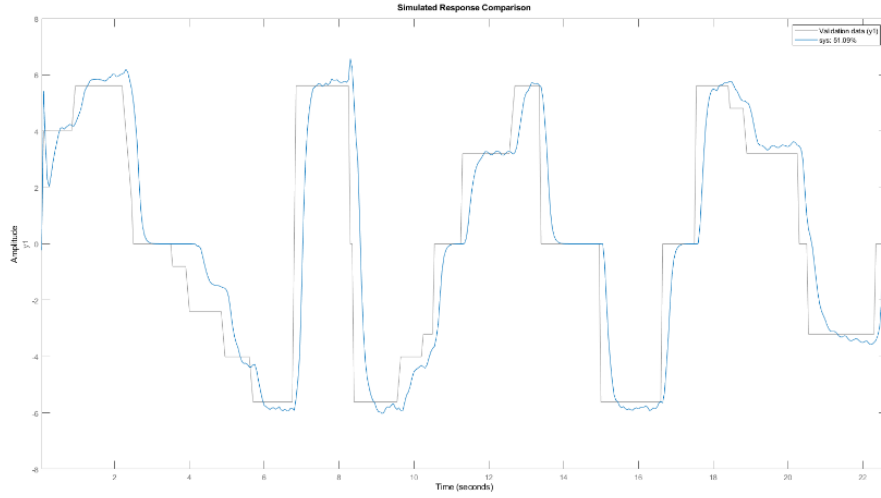


Figure 5.9. Sistem karşılaştırma sonucunu gösterir

5.2.6. Discussion

The transfer function derived from the system identification process represents the mathematical model that relates the input (desired input speeds) to the output (actual output speeds) of the robotic system. This model, which encapsulates significant information about the system's dynamics through its zeros, poles, and gain, serves as a foundational tool in understanding system stability, responsiveness, and dominant frequencies.

However, the transfer function reflects certain aspects of the system dynamics with a moderate fit of 51.09%. This level of fit indicates that the model captures just over half of the dynamics exhibited by the actual system data. Consequently, while the model provides a rough approximation of how the system behaves, it likely misses out on crucial dynamics, particularly non-linear behaviors not represented linearly.

The moderate fit can be attributed to several factors:

Sensor Limitations: The sensors used to collect data might have inherent limitations, such as noise, accuracy range, and resolution, which can affect the fidelity of the input data used for system identification.

Testing Conditions: The robotic system's operational environment during the testing phase may not perfectly replicate its intended operational setting. Variations in terrain, temperature, and other environmental factors can introduce discrepancies that are not accounted for in the lab or controlled environments.

Given these dynamics, and to address the non-linear components of the system behavior that a linear model might not capture, it is proposed to integrate a control strategy that employs both Fuzzy PID (FPID) and traditional PID control. FPID would be particularly advantageous in managing the non-linear aspects, leveraging fuzzy logic to handle ambiguity and imprecision in sensor readings and environmental conditions. In contrast, PID control would maintain robustness in linear scenarios, providing a well-established method for maintaining desired set points and stability.

This hybrid control approach is crucial for developing a more robust control strategy that aligns closely with the observed behaviors of the robotic system. By combining the strengths of both fuzzy logic and PID control, the system can dynamically adjust to a wide range of operating conditions, thereby enhancing performance and reliability. Designing a FPID controller involves integrating the principles of fuzzy logic with the classical PID control strategy. This hybrid approach enhances the controller's ability to handle uncertainty and nonlinearities in system dynamics, making it particularly useful for complex control problems where traditional PID controllers might struggle. Here's a detailed methodology to design an FPID controller:

5.3. Design controller

To control the motor by inputting the desired speed, the speed must be converted and interpreted, and the appropriate voltage signal provided to the motor. Figure 5.10. illustrates the block diagram of the control process. It consists of a speed-to-voltage conversion unit, which is measured using a linear equation. Then, a PWM signal is generated by the program and written to the driver circuit, which in turn drives the motor.

The desired speed is passed through a subtractor that outputs the error, which is the difference between the reference speed and the actual value measured by the sensor as feedback.

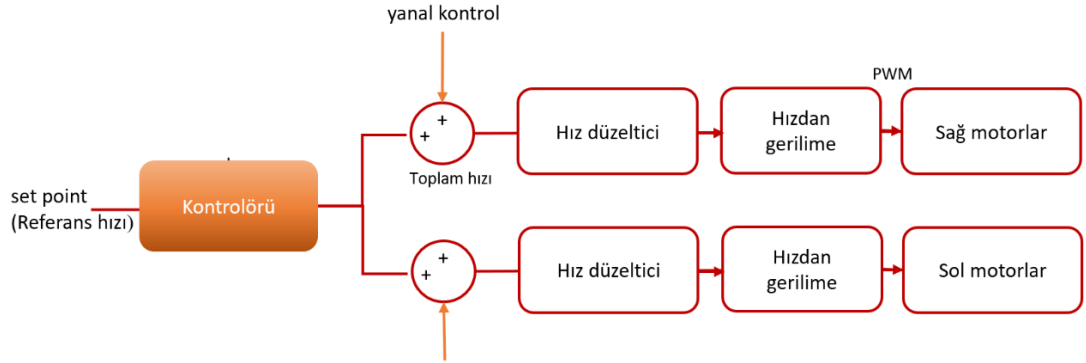


Figure 5.10. Illustrates the control system for managing the speed of left and right motors based on a reference speed set point. The system incorporates both lateral and speed correction.

This section details the design of three distinct controllers to compare their performance: a FLC, a PID controller, and a FPID controller. The FLC is developed using a Mamdani inference system with two inputs and one output, governed by 25 rules that determine the control actions. The PID controller, a widely used conventional control method, is designed to serve as a baseline for performance comparison. Finally, the FPID controller integrates fuzzy logic principles with the traditional PID control strategy, aiming to combine the strengths of both approaches. By evaluating and comparing these controllers, the study seeks to identify the most effective control strategy for the specific application, providing insights into the advantages and limitations of each method.

The controller design is based on angular velocity ω , but for robot control, the focus will be on linear velocity v . The relationship between angular and linear velocity is given by the following equation:

$$v = r \cdot \omega$$

where v is the linear velocity, r is the radius, and ω is the angular velocity.

5.4. Design FLC

The Mamdani inference system, is applied here to design a fuzzy controller with two inputs and one output. This system utilizes a rule-based approach to model the control behavior, where each rule is defined in terms of fuzzy logic.

The controller has two main inputs and one output:

1. Error (e)

The membership functions (MFs) for the input variable "e" consist of five distinct shapes: two trapezoidal MFs are positioned at the ends, and three triangular MFs are situated in between. These are configured based on the range of input values determined by the maximum angular speed the vehicle can achieve, which is 30 radians per second. However, the operational range for "e" is capped at 60 radians per second, calculated as the difference between the maximum (forward angular speed: 30 rad/s) and the minimum (backward angular speed -30 rad/s) possible values. To accommodate fluctuations around this operational range and ensure comprehensive coverage, the distribution was set to span from -60 to 60 radians per second. Consequently, the specific values chosen for this distribution are: [-60, -40, -20, 0, 20, 40, 60].

Table 5.1. Membership Function Parameters for Error Variable

MS name	Type	Range
NB	Trapezoidal	-80, -60, -40, -20
NS	Triangular	-40, -20, 0
ZE	Triangular	-20, 0, 20
PS	Triangular	0, 20, 40
PB	Trapezoidal	20, 40, 60, 80

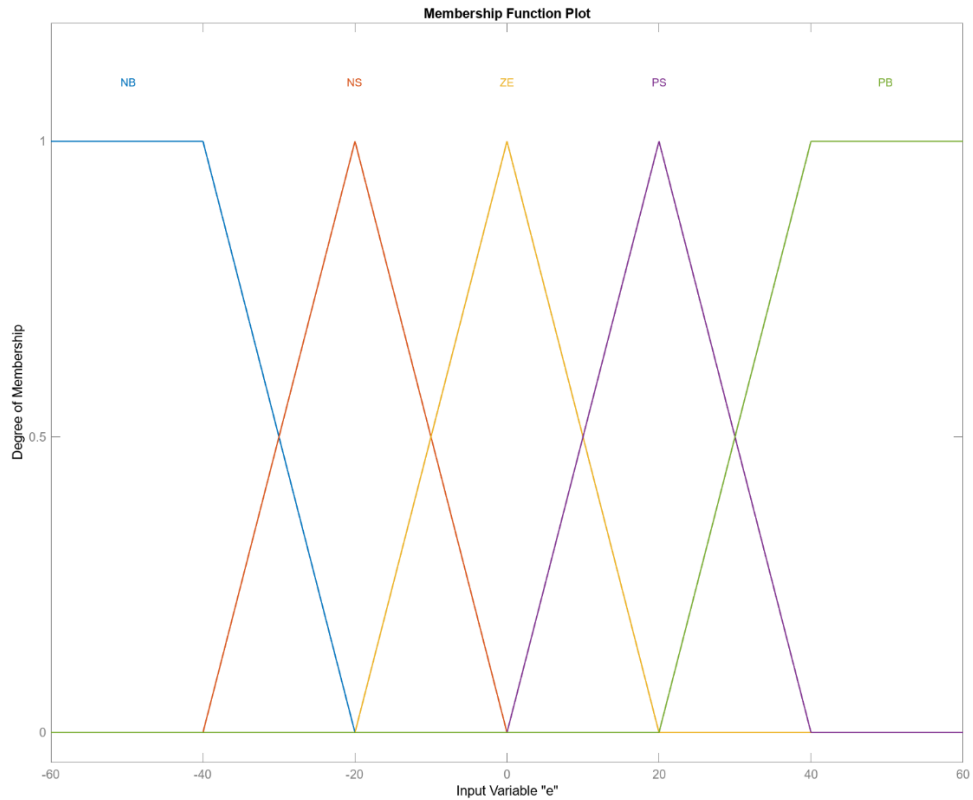


Figure 5.11. Membership Functions for Error

2. The derivative of the input u , denoted as u'

plays a crucial role in the control system design, ensuring satisfactory performance by adapting to changes in u . To determine the optimal range for u' , we examine the maximum rate of change in u , which occurs at the highest peak of the slope when the largest error difference is applied. The step pulse observed value of this slope is 100. Based on this measurement, the range for u' is appropriately set to encompass significant variations, leading to the selection of values at regular intervals within this range: $[-6000, -4000, -2000, 0, 2000, 4000, 6000]$. This setup is designed to effectively manage and respond to dynamic changes in the system, enhancing the controller's ability to maintain stability and performance under varying conditions.

Table 5.2. Membership Function Parameters for u'

MS name	Type	Range
NB	Trapezoidal	-8000, -6000, -4000, -2000
NS	Triangular	-4000, -2000, 0
ZE	Triangular	-2000, 0, 2000

PS	Triangular	0, 2000, 4000
PB	Trapezoidal	2000, 4000, 6000, 8000

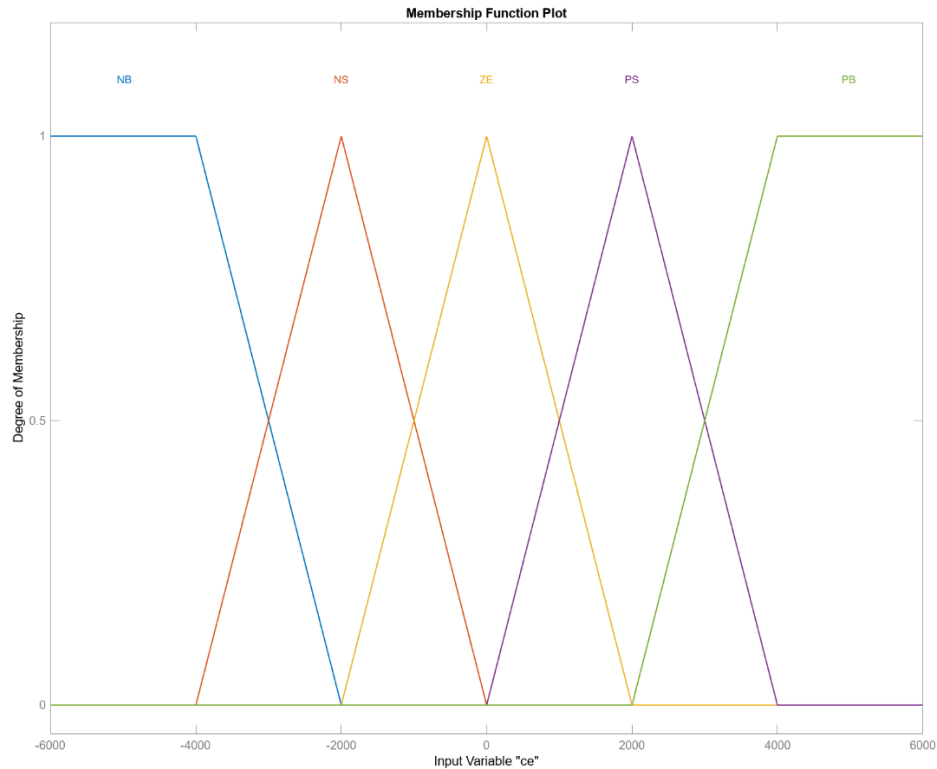


Figure 5.12. Membership Functions for derivative of the input

3. Output Variable:

The output also utilizes 5 MFs, but these are depicted with a different shape in your diagram, indicating a different method of handling output fuzzification compared to the inputs.

The output design was based on the system's requirements, maximum values of velocity. The selected output values are $[-30, -20, 0, 20, 30]$, chosen to meet the system's operational needs, ensuring robust and adaptive control across various conditions. This approach guarantees efficient and effective performance.

Table 5.3. Membership Function Parameters for Error Variable

MS name	Type	Range
NB	Trapezoidal	-40, -30, -20, -10
NS	Triangular	-20 -10, 0

ZE	Triangular	-10, 0, 10
PS	Triangular	0, 10, 20
PB	Trapezoidal	10, 20 30, 40

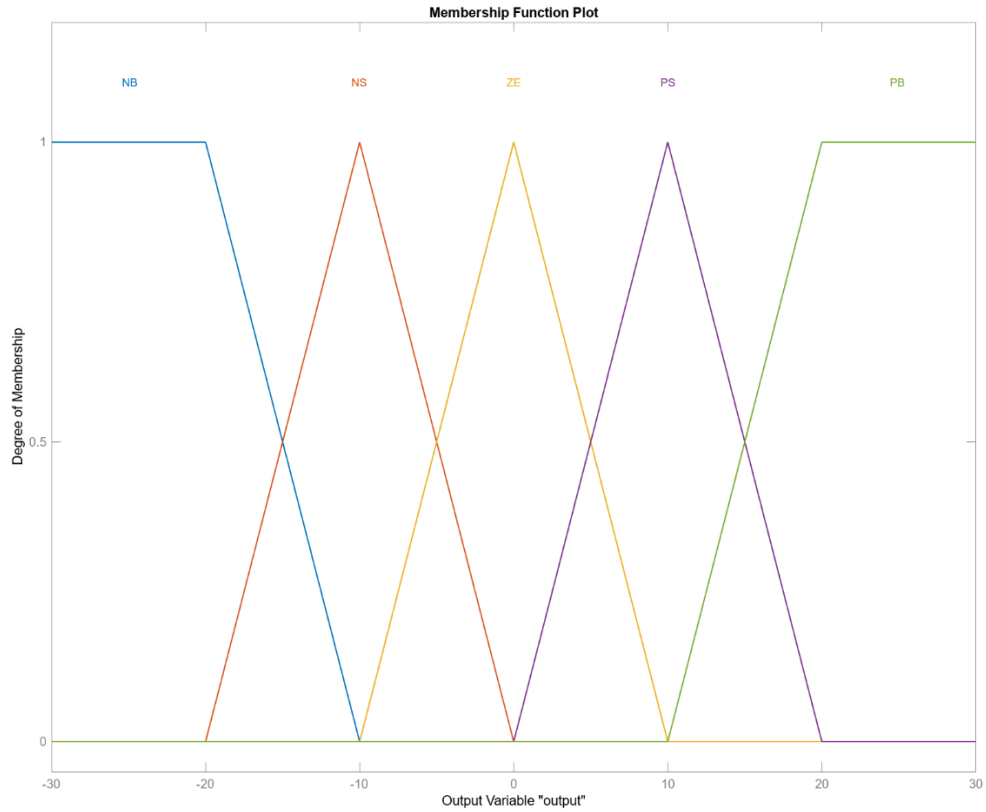


Figure 5.13. Membership Functions for output

Rule Base

The rule base consists of 25 rules that describe the relationship between the inputs and the output. The rules are structured as follows:

1. *If (e) is NB and (ce) is NB, then output is NB.*
2. *If (e) is NS and (ce) is NB, then output is NB.*
3. *If (e) is ZE and (ce) is NB, then output is NS.*
4. *If (e) is PS and (ce) is NB, then output is NS.*
5. *If (e) is PB and (ce) is NB, then output is ZE.*
6. *If (e) is NB and (ce) is NS, then output is NB.*
7. *If (e) is NS and (ce) is NS, then output is NS.*
8. *If (e) is ZE and (ce) is NS, then output is NS.*
9. *If (e) is PS and (ce) is NS, then output is ZE.*

10. If (e) is PB and (ce) is NS, then output is PS.
11. If (e) is NB and (ce) is ZE, then output is NB.
12. If (e) is NS and (ce) is ZE, then output is NS.
13. If (e) is ZE and (ce) is ZE, then output is ZE.
14. If (e) is PS and (ce) is ZE, then output is PS.
15. If (e) is PB and (ce) is ZE, then output is PB.
16. If (e) is NB and (ce) is PS, then output is NS.
17. If (e) is NS and (ce) is PS, then output is ZE.
18. If (e) is ZE and (ce) is PS, then output is PS.
19. If (e) is PS and (ce) is PS, then output is PS.
20. If (e) is PB and (ce) is PS, then output is PB.
21. If (e) is NB and (ce) is PB, then output is ZE.
22. If (e) is NS and (ce) is PB, then output is PS.
23. If (e) is ZE and (ce) is PB, then output is PB.
24. If (e) is PS and (ce) is PB, then output is PB.
25. If (e) is PB and (ce) is PB, then output is PB.

Table 5.1. The rule base is summarized in the following table.

	NB	NS	ZE	PS	PB
NB	NB	NB	NB	NS	ZE
NS	NB	NS	NS	ZE	PS
ZE	NB	NS	ZE	PS	PB
PS	NS	ZE	PS	PS	PB
PB	ZE	PS	PB	PB	PB

In Figure 5.14., The surface plot illustrates the output of the fuzzy inference system with respect to the two input variables, error (e) and change in error (ce). The plot shows how the system's output varies as a function of these inputs, providing a visual representation of the rule base's impact on the control action. The color gradient and contours depict different output levels, ranging from negative to positive values, indicating the system's response. This plot is crucial for understanding the relationship

between inputs and output, ensuring that the designed fuzzy controller performs as expected under various conditions.

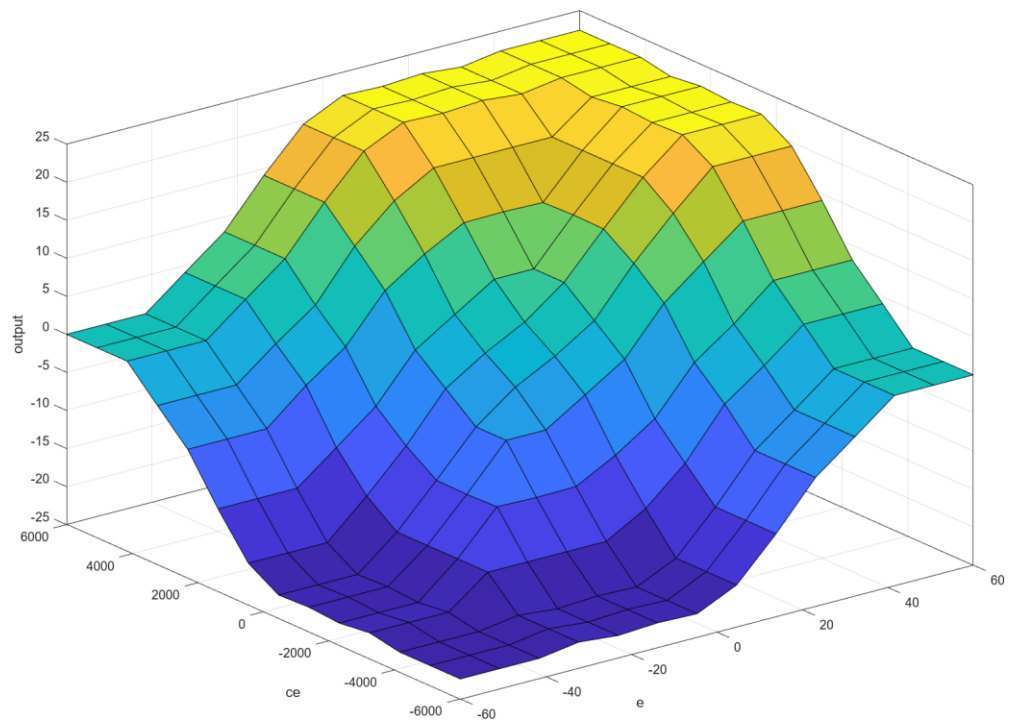


Figure 5.14. Surface Plot of the Fuzzy Inference System Output.

5.5. Modeling

Modeling is performed using Simulink in MATLAB, where three controllers are designed: PID, FLC, and FPID, for comparison purposes.

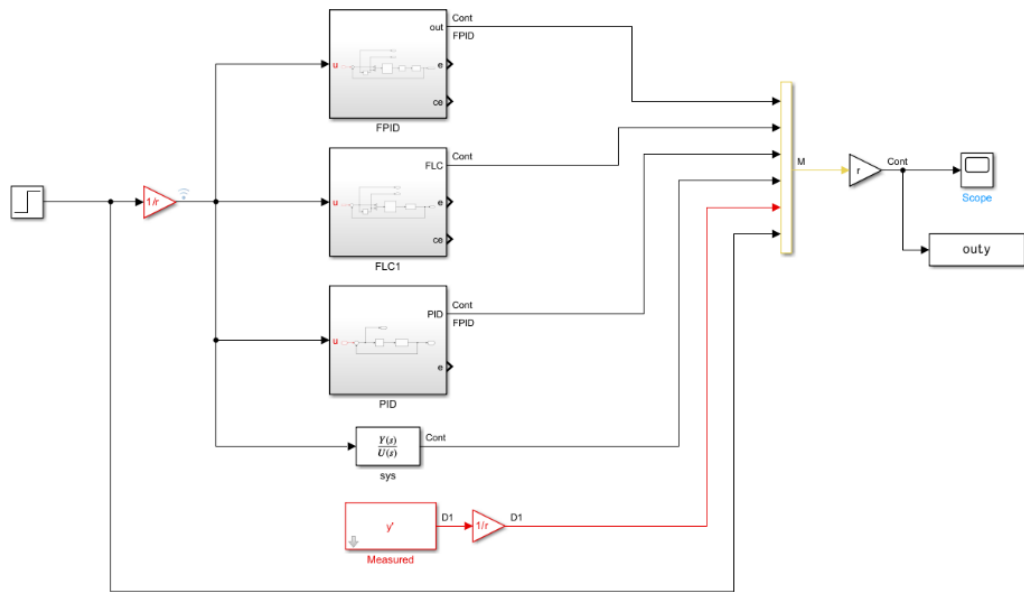


Figure 5.15. system modeling in Simulink.

The diagram illustrates a control system setup comparing three controllers: FPID, FLC, and PID. Each controller processes the input and generates an output speed. The responses of each controller are compared using a multiplexer (M) and visualized on a scope. This setup enables a direct performance comparison of the controllers under the same conditions.

5.5.1 Modeling and Design PID

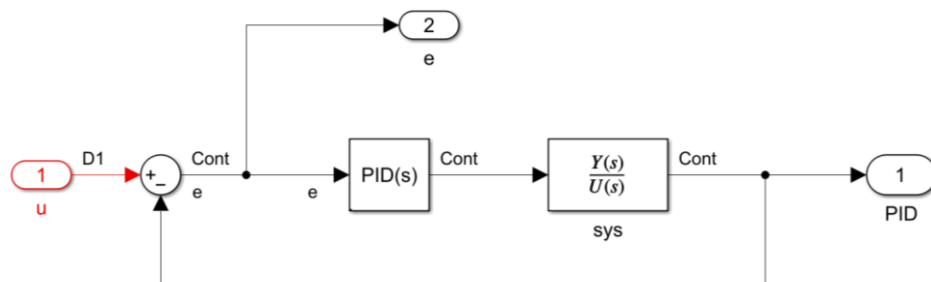


Figure 5.16. PID Controller Block Diagram

The block diagram represents a PID controller system. The input signal (u) is compared with the feedback signal, generating an error signal (e). This error is processed by the PID controller, which adjusts the control output to minimize the error.

In this specific PID controller design, the proportional gain K_p is set to 3, the integral gain K_i is set to 1, and the derivative gain K_d is set to 7. These parameter values are chosen to provide a balanced response, where the proportional term provides immediate error correction, the integral term eliminates steady-state error, and the derivative term helps in predicting and mitigating future error based on its rate of change.

K_i	K_p	K_d
75.4126601512025	4.85099122174835	0.0676996920115196

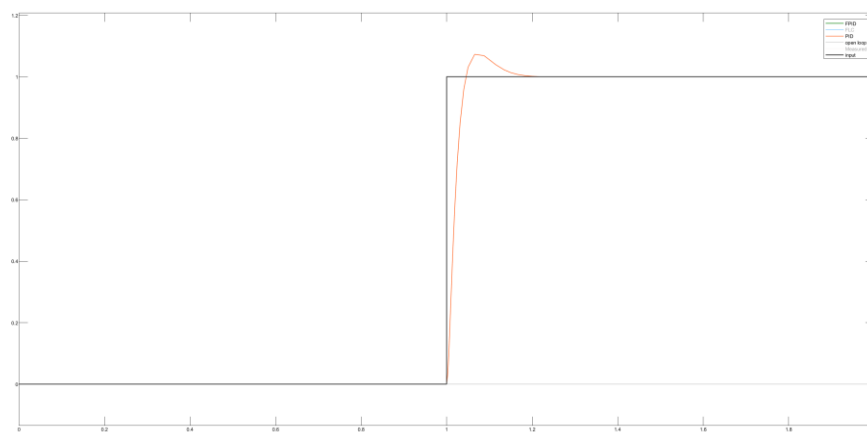


Figure 5.17. Step response of PID

Table 5.2. PID Controller Performance Metrics

PID Controller Performance Metrics	
<i>Rise Time_t</i>	40.030425s
<i>Overshoot</i>	7.2984%
<i>SettlingTime</i>	1.1386s

5.5.2 Modeling FLC

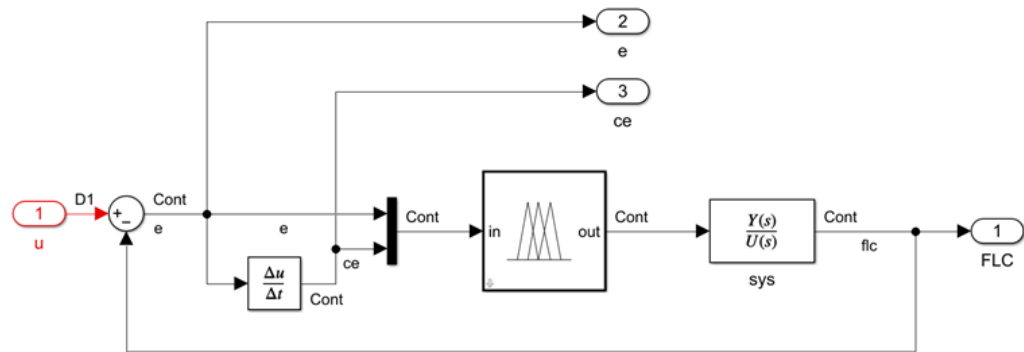


Figure 5.18. Block daigram of FLC system.

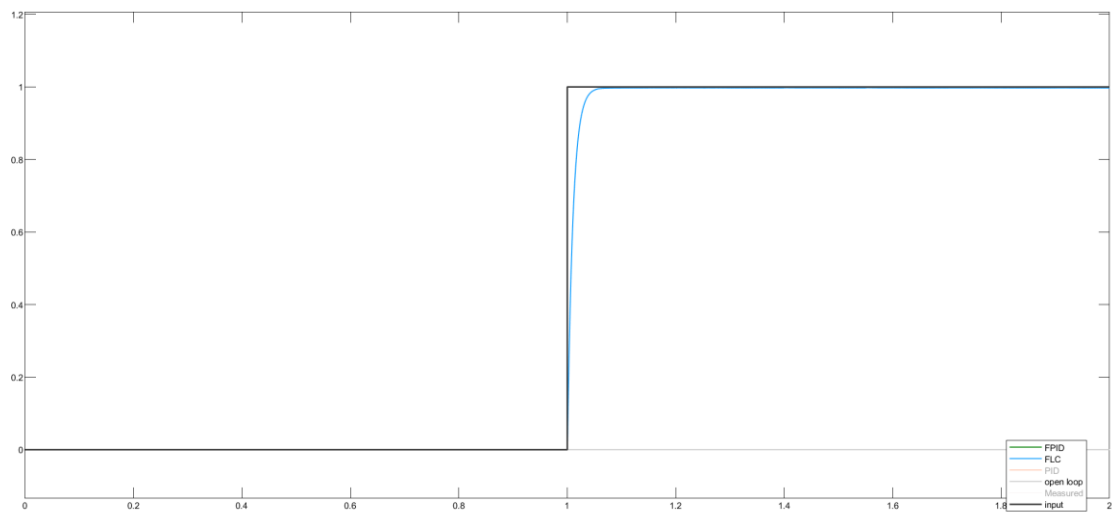


Figure 5.19. Step response of FLC

Table 5.3. FLC Performance Metrics

FLC Controller Performance Metrics	
Rise Time_i	0.021941 s
Overshoot	0.05631%
SettlingTime	1.039 s

5.5.3 Modeling and Design FPID

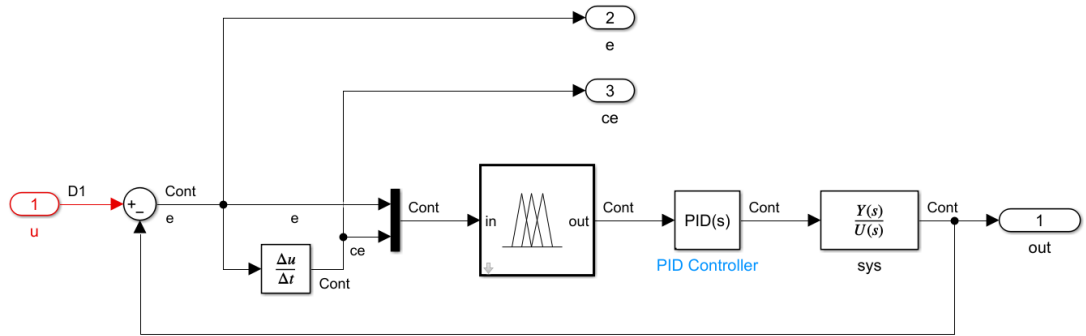


Figure 5.20. Block daigram of FLC system.

K_i	K_p	K_d
0.638159168922182	0.0543989149892157	0.00114859066969865

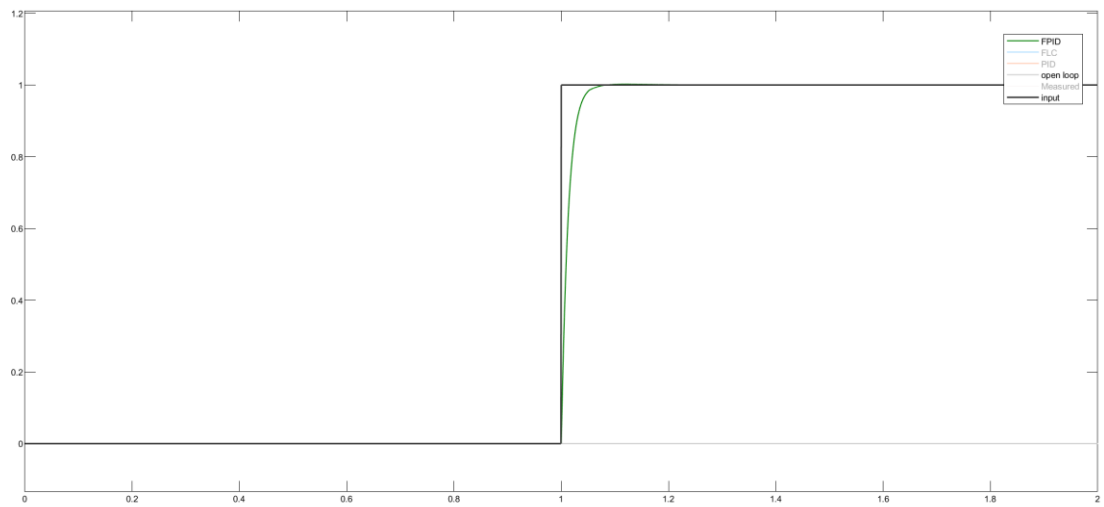


Figure 5.21. Step response of FLC

Table 5.4. FPID Performance Metrics

FPID Controller Performance Metrics	
Rise Time _i	0.027784s
Overshoot	0.226%
SettlingTime	1.0486s

5.6 Discussing result

In Figure 3, the signals for each of the controllers and the open loop system are displayed on top of each other for comparison and analysis. This method allows for the observation of differences in response and performance between the various controllers when subjected to the same signals.

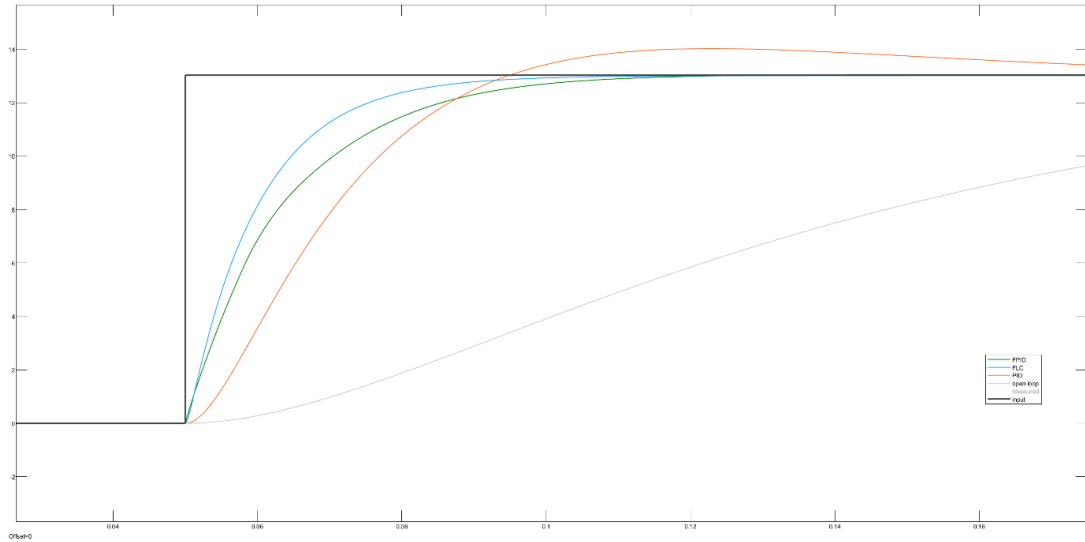


Figure 5.22. system response of open-loop, PID, FLC and FPID

	Rise Time	Overshoot	Settling Time	Absolute Error
FPID	0.029957	0	0.1015	4.2301
FLC	0.021564	0	0.089016	4.2349
PID	0.029807	7.6429	0.18832	4.0934
Open-loop	0.14751	0	0.27544	4.2356

5.6.1. Open Loop Response:

The open loop response, represented by the grey line, shows a slow and steady increase. Without any feedback control, the system exhibits a significant delay and does not reach the desired target value efficiently. This highlights the necessity for implementing a control mechanism.

5.6.2. PID Controller Response:

The PID controller, represented by the orange line, achieves the fastest rise time among the control strategies, quickly reaching the target output. However, it exhibits a noticeable overshoot before stabilizing at the desired value

5.6.3. FLC Response:

The FLC, represented by the blue line, exhibits the smoothest and most gradual approach to the target output. Although it has the slowest response time among the three control strategies, it avoids overshoot and provides a stable and robust response. The FLC's adaptive nature allows it to handle non-linearities and uncertainties in the system more effectively, making it suitable for complex and variable environments where maintaining stability is more critical than achieving rapid response.

5.6.4. FPID Response:

The FPID controller, shown by the green line, provides a more balanced and stable response compared to the PID controller. It reaches the target output with less overshoot and a slightly slower rise time. The fractional-order component of the FPID controller allows it to handle system dynamics more smoothly and effectively, offering a compromise between quick response and stability. This capability makes the FPID controller suitable for systems where reducing the risk of oscillations and achieving a steady response are critical. The FPID controller demonstrates improved robustness and adaptability, making it more effective in dealing with complex system behaviors and parameter variations.

While the FLC performs well for step changes, its slower response might be less effective in scenarios with larger or more rapid changes. This suggests that the FLC controller excels in maintaining stability but may lack the responsiveness required for significant variations in the system.

This behavior indicates that the FPID controller is effective in handling large step changes, adapting smoothly to achieve the target output with minimal oscillations. The fractional-order component of the FPID controller enhances its adaptability and robustness, allowing it to manage significant changes in the system more effectively.

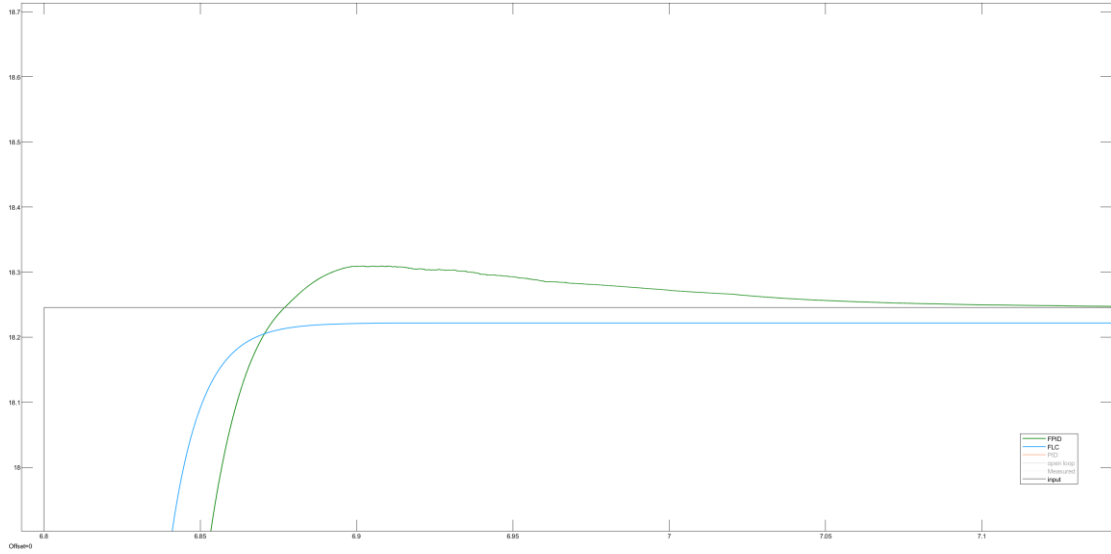


Figure 5.23. The controller's response when going up appears significant.

The FLC controller continues parallel to the reference value, while the FPID controller is completely stable.

5.7 Conclusion

On the other hand, the FPID controller stands out for its balanced approach, combining the advantages of quick response and enhanced stability. Its ability to handle large step changes and adapt smoothly to achieve the target output with minimal oscillations makes it particularly effective. The fractional-order component of the FPID controller enhances its adaptability and robustness, allowing it to manage significant changes in the system more effectively.

Overall, the FPID controller emerges as a highly effective solution, offering a desirable compromise between response speed and stability. Its ability to manage system dynamics smoothly and its robustness to parameter variations make it a superior choice for applications requiring both efficiency and reliability.

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ÖZGEÇMİŞ

Khaled HAMIDI, Karabük Üniversitesi Mechatronik Mühendisliği öğrencisidir. Bölüm birinciliği ile mezun olmuş ve tüm derslerini üç yıl içinde tamamlamıştır. 2022 yılında ilk defa düzenlenen Karabük Üniversitesi Hackathon'una katılarak ödülü kazanmıştır. TÜBİTAK 2209A Araştırma Programı kapsamında bir araştırma projesinde yer almakta olup, dört boyutlu robotların kontrol algoritmaları üzerine bir araştırma makalesi yayınlamıştır. ICENSOS konferansında, "SABİT GRANÜL MİKTARLARI İÇİN DOLUM MAKİNESİ TASARIMI VE İMALATI" projesini sunmuştur. İbni-Firnas Kulübü'nün kurulmasına katkıda bulunmuş ve Teknofest'e katılmıştır. Ayrıca, mühendislik öğrencilerinin yeteneklerini geliştirmek amacıyla Robotech Eğitim Takımı'nı kurmuştur.